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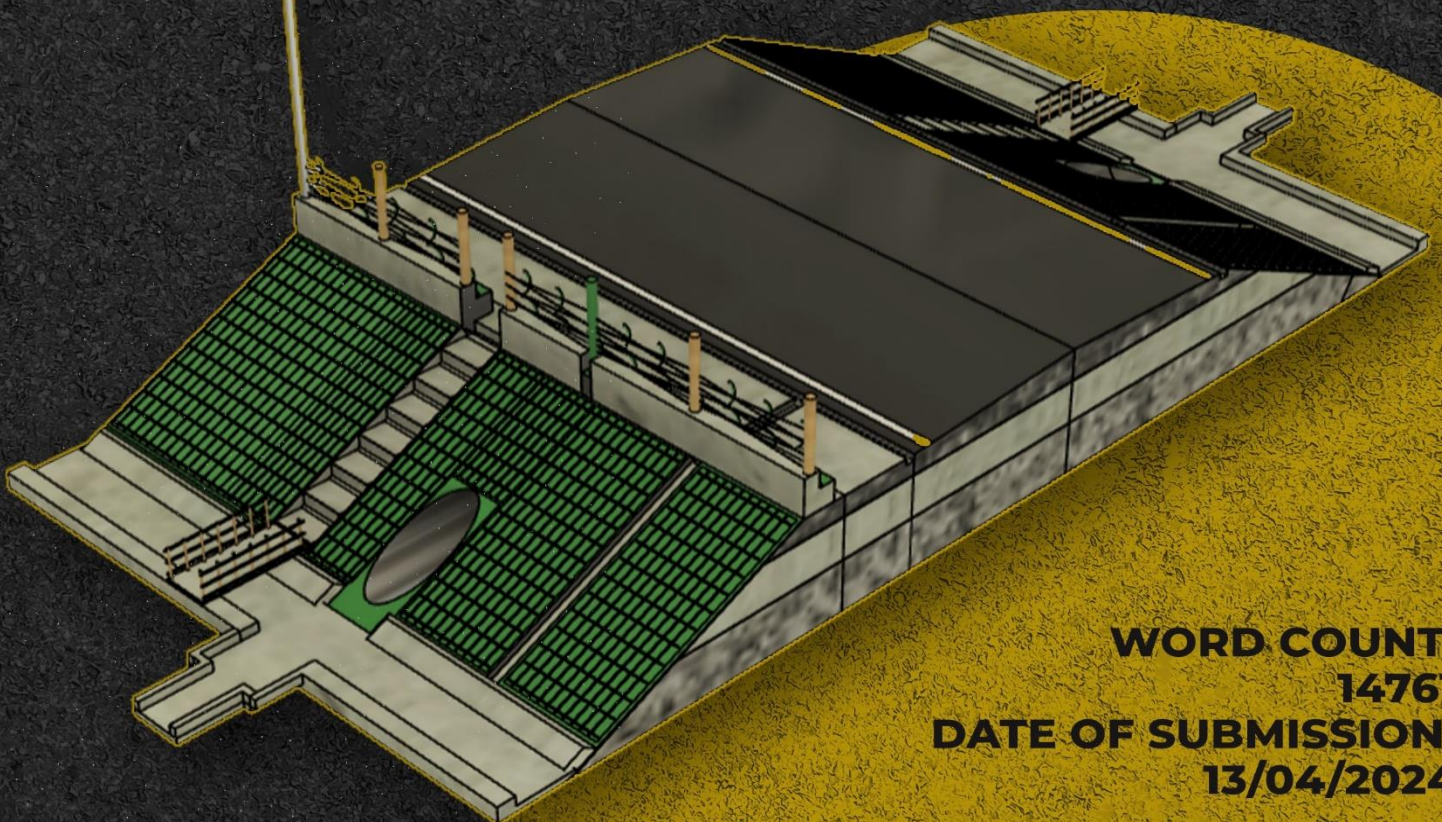
**ENGINEERING FOUNDATION: PRINCIPLES, DESIGN  
AND COMMUNICATION**

**INDE1001**

**2024 EWB CHALLENGE**

**IMPROVED LOCAL ROAD ACCESS  
IN THE WET SEASON IN SAIBAI ISLAND**

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# Executive Summary

Engineers Without Borders (EWB), Australia, in partnership with the Torres Strait Island Regional Council, have introduced their 2024 EWB Challenge to find a solution for a key problem faced by the Saibai Islanders. Saibai is an extremely flat mud mangrove island. The islanders are living in a narrow strip of beachfront, which is only a few meters above sea level. Irrespective of the stone seawall constructed in the mid-1900's, this area gets inundated several times a year by King tides. Inundation and inaccessibility of the local road during high tide is one of the key problems that directly affect the whole community, including school children. This hinders the children from attending school regularly due to floods and muddy roads.

Our project proposes a durable and sustainable solution to this pressing mobility problem. The key steps of the methodology included the identification of key stakeholders, identification of client requirements through stakeholder consultations while identifying challenges and limitations. Generation of design alternatives and the selection of the best option was carried out through a comprehensive evaluation considering environmental, economic and social considerations.

The local Saibai community, local government, constructors/suppliers, and investors were the key stakeholders of this project. The Saibai community is the main stakeholder who has the highest interest in this road project due to the direct impacts on them. The Saibai community needs a safe and durable solution that can withstand the wet season. As a stakeholder, the local government has a higher control since they can influence the project as the relevant authority. Their main interest is having sustainable infrastructure that aligns with the community's needs. Investors are the next major stakeholder with a high interest in the financial return and have the power to implement the project. Finally, the constructors and suppliers as stakeholders have relatively lower power and interest. They are interested in the timely completion of the project following the quality standards, with targeted profits and a safe working environment.

The research phase of this project was the most challenging due to various limitations. The inability of the team to visit the Island site and have direct contact with the stakeholders was the main limiting aspect. Therefore, the research was done using materials provided by EWB and other resources available on the Internet. As a result, the use of some outdated sources was inevitable.

The proposed sustainable solution comprised of an elevated road structure made of locally available material, facilitating mobility for the Saibai Islanders during the wet season. The pavement structure includes layers such as riprap, stabilized soil, reclaimed asphalt pavement, and micro-surfacing, each serving specific purposes in the geotechnical properties such as bearing capacity while enhancing resilience against wear. The road's embankment is equipped with erosion control measures using geomats to safeguard against erosion and stabilization. Drainage systems, including roadside drains and culverts, manage water runoff effectively to prevent pavement flooding during heavy rains. Concrete pavement ensures the durability and safety of pedestrian walkways. The pedestrian walkway is complemented by a green wall featuring live bamboo and hydrogel-infused soil as a further measure for erosion control, providing many other environmental benefits and supporting habitats for wildlife. This innovative addition not only enhances environmental resilience but also offers aesthetic benefits and promotes a healthier urban environment for Saibai Island residents.

Sustainability, safety, durability, and local material use are the four design criteria used in this proposed solution, with 20%, 30%, 30% and 20%, respective weightages. Since this is a road construction project, durability and safety were highly significant. The road was designed to withstand adverse climatic conditions while ensuring the safety of the road users and the community around them. Material selection considerations focused on lowering the cost by prioritising locally sourced materials or having a higher recycled content. This would also contribute to minimum environmental damage and enhanced sustainability.

Each design criteria selected had its own constraints. Finding locally sourced yet durable materials with recycled content was challenging in large-scale construction such as this. Adherence to a strict budget that doesn't compromise on implementing a sustainable solution prioritizing the preservation of Saibai Island's natural habitat was challenging. Lastly, the durability concerns, focusing on slope stability and vegetation management, ensuring the longevity of the infrastructure necessitated innovative approaches.

This project will ultimately benefit the Saibai community as it will give them mobility through a properly functioning road to travel in extreme weather conditions. The design proposed in this project can also be used in other communities with a similar background.

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## 1.0) Introduction

The 2024 EWB Challenge was launched by Engineers Without Borders (EWB), Australia in collaboration with the Torres Strait Island Regional Council to address a major issue that the people of the Saibai Islands are now facing. The mud mangrove island of Saibai is incredibly flat. Only a few meters above sea level, the islanders reside in a slender stretch of seashore. Despite the construction of a stone bulwark in the mid-1900s, they are frequently submerged by King tides every year. One of the main issues that immediately affects the whole community, especially schoolchildren, is inundation and the local road's inaccessibility during high tide. Floods and muddy roads make it difficult for the kids to routinely attend school.

Our aim is to suggest a solution to one of the island's most pressing issues: developing a proposal to improve road access during the rainy season when heavy rains and flooding make roads muddy and inaccessible. The proposed concept focuses on creating roads that are safe for all, including school children, even in adverse weather conditions. Additionally, the solution prioritizes durability, sustainable use of construction materials and ease of maintenance, empowering the local community to manage and maintain the roads effectively. An essential aspect of the concept design includes implementing a proper drainage system to mitigate flooding and ensure efficient rainwater drainage, reducing the risk of natural disasters such as flooding (EWB Challenge | Design Area 3 – Infrastructure, n.d.).

To explore the design options, extensive research was conducted on Saibai Island's history, present infrastructure and related issues, geography, culture, economy, and political background. Understanding these factors enabled us to establish connections between the proposed concept and its potential impact on various aspects of the island, such as the economy and road infrastructure. There were many parties who were interested in this project. Recognizing the diverse interests of the four major stakeholders of this project, each with distinct needs and desires, was important. The concept design has been tailored to effectively meet the requirements of these stakeholders.

The resulting conceptual solution, detailed in this report, includes design concept drawings, justification of design aspects, and an approximate budget. The model drawings, specifications, and the physical model, would allow understanding of the design concept clearly. By comprehensively addressing the challenges associated with building access roads to improve mobility during wet



season, our solution aims to enhance the livelihoods of Saibai Island's residents and contribute positively to the island's development.

## **2.0) Project Background**

To understand the scope and required solution of the project, various factors were studied under this section: climate, existing road infrastructure, culture, education, demographics, geography and environment, political background, economic implications of the project and materials are the factors discussed.

### **2.1) Climate**

It was vital to know the background information on the climate of Saibai Island to determine optimum design solutions. This section will mainly focus on the wet season as our topic is mainly based on the wet season.

Saibai Island experiences a tropical climate characterized by a distinct rainy season, typically spanning from October to April, with heavy rainfall and high winds at its onset. While the island isn't frequently hit by cyclones, tidal surges from cyclones along the eastern Australian coast indirectly affect it, along with king tides occurring several times a (EWB Australia, 2024, p. 8). Despite the presence of a seawall, the coastal area remains vulnerable to flooding during high tides, a situation expected to worsen due to climate change (RPS Group, 2011). The wet season accounts for about 95% of the annual rainfall, with El Niño bringing below-average rainfall and La Niña resulting in above-average rainfall (EWB Australia, 2024, p. 8). Even though the rainy season of Saibai Island deposits around 2m of rain, in the dry season, the island goes short of water (Tagai State College, n.d.).

### **2.2) Existing Road Infrastructure**

The background information on the existing road infrastructure is crucial for designing a solution that suits the current situation. Nature, the roads, the modes of transportation and types of vehicles mainly used were studied under this section.

### 2.2.1) Nature of the Roads

In Saibai, most roads are made of 100 mm thick fiber-reinforced concrete on top of a stabilized subgrade. Many of the roads are 4 meters wide and badly cracked due to various reasons like lack of maintenance and bad climate. The lack of maintenance results from the high expenditure required, which is not affordable for the island. (RPS Group, 2011, p. 128)

Furthermore, unsealed or gravel roads can also be seen on the island, which becomes even more dangerous than concrete-reinforced roads during floods.

Figure 1 shows an unsealed or gravel road on Saibai Island. Whether the initial road construction was a concrete one or not is not clear. However, the road has now come to a level that is highly prone to getting muddy during the wet season. From the visible nature, it can also be assumed that even regular rain could make this road dangerous.

The gravel road shown in Figure 1, next to Saibai Island Primary Health Care Centre highlights the importance of paying more attention to improving these kinds of roads.



*Figure 1: Depiction of the Google Street view of an unsealed or gravel road located next to Saibai Island Primary Health Care Centre. Taken by Ti Cooper in 2024.*

### 2.2.2) Road Users

Three parties mainly use the internal roads of Saibai Island.

They are:

1. The locals and settlers for their daily activities like occupations, etc.
2. School students to attend their education.
3. People associated with the construction industry in development projects, maintenance, etc.

Besides the above needs, Islanders commonly used vehicles for general and emergency medical services.

According to the Australian Bureau of Statistics, at the census date (2021), only 22 people have used a car to travel to work (both as a passenger and a driver). However, according to the 2006 Census, there were more than 79 privately owned vehicles on the Island. (RPS Group, 2011, p. 128) Due to the data disparity, we cannot exactly conclude on the number of vehicles being used regularly. However, it can be concluded that a considerable number of vehicles are still being used.

Furthermore, considering the remoteness, expensiveness of fuels, and proximity of town amenities, we can assume that most islanders tend to take short excursions.

### 2.2.3) Types of Vehicles Used

The types of vehicles used depend on the purpose it is used for. Compact, middle-sized, and economical cars are used on the island for general purposes.

Figure 2 depicts some cars parked on the Island, captured in Google Street Views.

More heavy and intense vehicles can be seen on the Island for the construction industry. Figure 3 depicts a few of such vehicles captured in Google Street Views.

Furthermore, it can be assumed that the heavy-duty vehicles used in the construction industry are not used during the wet season because of heavy rains and muddy situations. Thus, more focus should be given to the vehicles that are generally used rather than the construction industry in our design process.



*Figure 2: Depiction of the Google Street view of 2 cars parked on the Island. Taken by Ti Cooper in 2024.*



*Figure 3: Depiction of the Google Street view of some heavy vehicles used in the construction industry. Taken by Ti Cooper in 2024.*

## 2.3) Culture

Saibai is an island with a rich culture dating back thousands of years back. They value their culture immensely because it is a part of their way of life. When introducing a new infrastructure like a road, the designing company should highly focus on the feasibility of the design to that of the culture of the Saibai community. Thus, briefly understanding their culture is vital to build up a sustainable solution that could be practically implemented.

Culture is diverse in Saibai, it may come in the forms of art, songs and unique dances. For example, if we consider songs, they are not only a part of their culture, but it also comprises historical events that affected the life of Saibai islanders (Harvey, 2020). If the the design of the road included some aspects which may offend the Islanders, then it will negatively affect the implementation of the project. As a result, high attention should be given in designing process to secure and promote the island's cultural practices, religious traditions, and community togetherness.

## 2.4) Education

It has been identified that school students are one of the highly affected categories during the flooding in the wet season. Therefore, researching their background is important in finding an appropriate solution for the road project.

From figure 4 (Australian Bureau of Statistics [ABS], 2021b), it can be identified that more than 63 per cent of the students are attending primary school. Thus, it is evident that the road design should consider the safety and needs of primary school students.

| Type of educational institution attending<br>Aboriginal and/or Torres Strait Islander people attending an educational institution | Saibai Island | %    | Queensland | %    | Australia | %    |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------|------|------------|------|-----------|------|
| Preschool                                                                                                                         | 4             | 3.1  | 6,070      | 6.6  | 25,485    | 8.0  |
| Primary                                                                                                                           | 83            | 63.4 | 35,874     | 38.9 | 120,864   | 37.9 |
| Secondary                                                                                                                         | 17            | 13.0 | 26,207     | 28.4 | 83,631    | 26.2 |
| Tertiary - Vocational education (including TAFE and private training providers)                                                   | 3             | 2.3  | 4,967      | 5.4  | 21,273    | 6.7  |
| Tertiary - University or higher education                                                                                         | 0             | 0.0  | 6,362      | 6.9  | 21,149    | 6.6  |
| Other                                                                                                                             | 0             | 0.0  | 2,803      | 3.0  | 7,310     | 2.3  |
| Not stated                                                                                                                        | 15            | 11.5 | 9,970      | 10.8 | 38,814    | 12.2 |

Table 1: Type of educational institution attending (ABS, 2021b)

From figure 5 (ABS, 2021b), it can be seen that the majority, 25 per cent of the population, has an education level till year 12, and there is no person with a bachelor's degree level or above. Thus, in the aspect of maintenance of the road, less technical applications should be included so that it can be easily maintained by the locals.

| Level of highest educational attainment<br>Aboriginal and/or Torres Strait Islander people aged 15 years and over | Saibai Island | %    | Queensland | %    | Australia | %    |
|-------------------------------------------------------------------------------------------------------------------|---------------|------|------------|------|-----------|------|
| Bachelor Degree level and above                                                                                   | 0             | 0.0  | 11,028     | 7.0  | 40,554    | 7.4  |
| Advanced Diploma and Diploma level                                                                                | 17            | 10.1 | 10,614     | 6.8  | 34,681    | 6.3  |
| Certificate level IV                                                                                              | 3             | 1.8  | 6,227      | 4.0  | 23,374    | 4.3  |
| Certificate level III                                                                                             | 22            | 13.1 | 27,557     | 17.5 | 84,821    | 15.5 |
| Year 12                                                                                                           | 43            | 25.6 | 28,526     | 18.2 | 82,998    | 15.2 |
| Year 11                                                                                                           | 10            | 6.0  | 11,349     | 7.2  | 43,335    | 7.9  |
| Year 10                                                                                                           | 16            | 9.5  | 28,737     | 18.3 | 102,707   | 18.8 |
| Certificate level II                                                                                              | 0             | 0.0  | 344        | 0.2  | 1,262     | 0.2  |
| Certificate level I                                                                                               | 0             | 0.0  | 52         | 0.0  | 241       | 0.0  |
| Year 9 or below                                                                                                   | 21            | 12.5 | 18,908     | 12.0 | 75,326    | 13.8 |
| Inadequately described                                                                                            | 21            | 12.5 | 3,131      | 2.0  | 11,038    | 2.0  |
| No educational attainment                                                                                         | 0             | 0.0  | 1,193      | 0.8  | 5,242     | 1.0  |
| Not stated                                                                                                        | 20            | 11.9 | 9,409      | 6.0  | 41,467    | 7.6  |

Table 2: Level of highest educational attainment (ABS, 2021b)

## 2.5) Demographics

In creating a more tailored road solution, it is vital to know the characteristics of the population of Saibai Island since they are the customers who will be using the road. The total population of this island is about 340, with men taking up 53.7% and women taking 46.3% while there are around 69 families, with the average number of people in a household being 3.6 (ABS, 2021a).

After studying the table below containing data about the amount of people by age group, it is evident to us that the largest age group in Saibai is children from age 5-9 years, which accounts for 17.1 per cent of the whole population. Thus, the importance of considering a child-friendly design for pedestrian paths is highlighted.



Table 3: The number of people in each age group (ABS, 2021a)

| Age               | Population | Percentage |
|-------------------|------------|------------|
| Median age        | 25         | N/A        |
| 0-4 years         | 30         | 8.8        |
| 5-9 years         | 58         | 17.1       |
| 10-14 years       | 41         | 12.1       |
| 15-19 years       | 27         | 7.9        |
| 20-24 years       | 15         | 4.4        |
| 25-29 years       | 29         | 8.5        |
| 30-34 years       | 28         | 8.2        |
| 35-39 years       | 23         | 6.8        |
| 40-44 years       | 13         | 3.8        |
| 45-49 years       | 16         | 4.7        |
| 50-54 years       | 8          | 2.4        |
| 55-59 years       | 17         | 5.0        |
| 60-64 years       | 13         | 3.8        |
| 65-69 years       | 8          | 2.4        |
| 70-74 years       | 9          | 2.6        |
| 75-79 years       | 0          | 0.0        |
| 80-84 years       | 0          | 0.0        |
| 85 years and over | 5          | 1.5        |

## 2.6) Geography and Environment

Saibai is a big, low-lying island about 4 kilometers from the Papua New Guinea mainland. Uninhabited Kauamag Island is located close to the north and is separated from Saibai by a 7-kilometer-long waterway that is between 180 and 650 meters wide and nearly obstructed at the east end. The highest point of the island is 1.7 m above sea level. Saibai Island is a flat Island and primarily a mangrove island. The island is about 20km by 15km but only a small part is inhabited (*Torres Strait Island Regional Council [TSIRC], n.d.-c*).



Figure 4: Satellite view of Saibai Island and nearby islands(Tageo, n.d.).

There are 2 key environmental assets in Saibai Island. Namely, they are a high-quality mangrove habitat and a lagoon functioning as a home for many endemic and rare fauna. Floral life and fauna found in Saibai Island are key factors in Saibai Island. It is an island for many endangered plants (RSP Group, 2011). TSIRC is very concerned about the protection of the floral life and fauna on this Island. As we are planning and carrying out this project we should adhere to sustainable factors like air pollution, emission reduction, sustainable construction, etc. and prevent the impact that will happen to the floral life and fauna.

## 2.7) Political Background

The Torres Strait Island Regional Council (TSIRC) is a local government authority responsible for providing local government services to the Torres Strait Islands. Saibai Island is one of the Torres Strait islands. This council is in charge of the development of the community, local infrastructure, and environmental management of the island. This council was formed on March 15, 2008, following elections under the Local Government Act 1993. This was formed by abolishing the 15 Torres Strait Island governments and establishing the Torres Strait Island Regional Council under a proposal by the Local Government Reform Commission in 2007 (EWB Australia, n.d.-b). Since then, the TSIRC has been critical in regional governance and administration. Improving locally accessed roads in the wet season can be considered under the development of infrastructure in Saibai Island, TSIRC is directly concerned because the development of the infrastructure for the use of future generations on the island is one of their main functions. Considering the sustainability factors and protecting the cultural heritage of the island when planning the development will be preferred by the local governing council, TSIRC.

## **2.8) Economic Implications of the Project**

Enhanced road access on Saibai Island is vital for economic sustainability and self-determination. Opportunities like aquaculture, tourism, agriculture, and infrastructure development are key for growth. Improved roads facilitate trade, boost tourism, enhance agricultural productivity, and create jobs. They also attract businesses, increase property values, and improve emergency services. Despite initial investments, long-term cost savings make the project economically viable.

## **2.9) Materials**

Obtaining information on the materials available on the island and materials that may have potential use in the project is vital to build up a sustainable solution that is both durable and cost-effective.

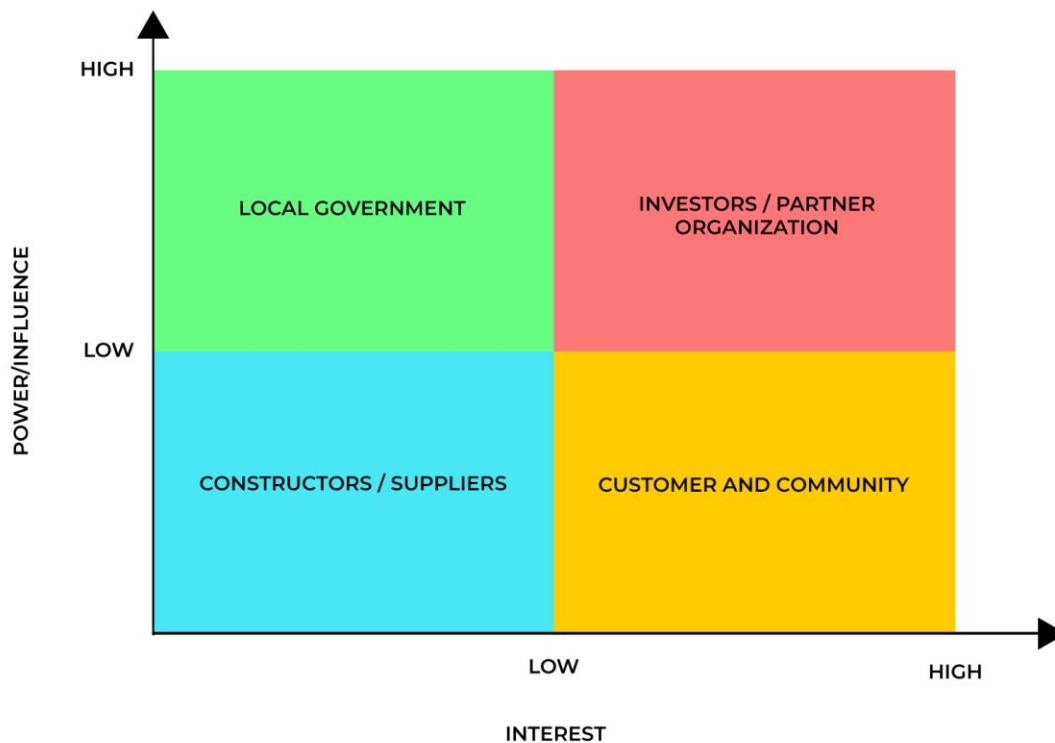
Saibai is a flat island containing many mangrove habitats and high-quality (RPS Group, 2011, p. 8). The vegetation mainly includes mango and fig trees, coconut, coral platforms, and swamps (RPS Group, 2011, p. 21). Furthermore, many thin vine forest bands are at the edge of the mangrove forest (RPS Group, 2011, p. 35).

There are certain areas where we cannot get materials since the ecosystem and the biodiversity in such places need to be protected, such places include Savannah grasslands to the south of the village, Extensive wetland and mangrove areas and Coastal mudflat areas along the shoreline of the island (RPS Group, 2011, p. 39).

The island contains a lot of acid sulphate soil. The soil contains many iron sulphides (mostly pyrite). When exposed to water and oxygen, this soil can have several types of iron compounds and sulphuric acid. It can produce some heavy metals through a chemical reaction that occurs naturally or can be sped up artificially. However, the soil is toxic and could cause corrosion to buildings and other infrastructure (RPS Group, 2011, p. 63).

### 3.0) Stakeholders

There can be many stakeholders such as suppliers, creditors, customers, trade unions, media, community, investors, government and employees, relevant to the project. Among these, four key stakeholders were selected according to their power and interests toward our project. The powers and interests of those chosen stakeholders are displayed in a chart below.



*Figure 5: Power influence vs interest of the stakeholders in the project*

### 3.1) Investors / Partner Organization

#### 3.1.1) Overview

Investors are among the key stakeholders in our project to improve local road access during the wet season on Saibai Island. Since developing local roads is highly expensive, a large-scale community project like this would not be practically implemented if it were not for the funds provided by the investors.

The Engineers Without Borders (EWB) Challenge- 2024 is officially sponsored by the “RS Group” (EWB Australia, n.d.-c). Therefore, that company will be considered the investor or partner organization for this project.

The RS Group provides various products and service solutions ranging from designing to operating industrial equipment. Furthermore, it also has a major consideration on the safety and sustainability of their work (RS Group, n.d.).

### 3.1.2) Concerns

The investor's main concern will be the trustworthiness of the project plan supported by scientific research-backed evidence, as well as the team behind it. Truthfulness, professionalism, ethical conduct, and honesty will be major components that are expected by the investors throughout the process since they do not like their invested money being wasted in the event of failure of the project due to lack of mentioned qualities.

Furthermore, the investors are highly concerned about the impact of this project on other stakeholders. They do not want any bad reputation for their name due to the project's negative impacts, which will indirectly affect their net profit.

### 3.1.3) Needs

#### **Profit**

Profit is the main goal of investors; this may be achieved directly through income from certain projects or indirectly through activities related to Corporate Social Responsibility (CSR), which helps to improve the company's reputation. By supporting community welfare and adhering to global sustainability trends, businesses may draw in customers who appreciate their commitment to sustainability without expecting quick financial benefits. This initiative, which falls under the indirect profit model, attempts to develop local infrastructure by enhancing the investor's reputation and the community's goodwill.

#### **Successful completion and implementation of the project**

The method of business communication in the corporate world is mainly through reports like ESG reports and official media publications. The minimum requirement for the business to publish the achievement gained via the CSR project is the successful completion and implementation. Without it, the investment company lacks evidence for the publication.



### 3.1.3) Wants

#### **High Community Satisfaction**

As a CSR project, the high satisfaction of the main stakeholder, which is the customer (in this case, the Saibai community), is vital. It comes under the “wants” category since the investment company can publish their already finished project with even average or neutral community satisfaction.

However, as investors who are concerned not only with monetary benefits but also the benefit of the community, the target is to achieve maximum satisfaction, which is sustainable in the long term.

### 3.1.4) Impacts on Investors If a Solution Was Not Found

#### **Short -Term**

If a solution was not found for the project, one of the positive short-term impacts on the investors would be having the opportunity to select another community development project that aligns with their goals and budget. The new project also must have the potential to make a positive impact on the community which ultimately will increase their good reputation.

One of the negative short-term impacts on the investors, if no solution is found for the project, would be in the scenario of not having another suitable project to select by the investors that aligns with their goals and budget.

#### **Long-Term**

In the long run, the effects are primarily detrimental. A long-term economic downturn in the area might be caused by persistently bad road conditions, which would then affect property values and the viability of new developments. The lack of access to emergency services will raise the dangers to public health and safety, and lower local educational attainment might lower the caliber of the workforce and indirectly effect investment decisions.

### 3.1.4) Impacts on Investors If a Solution Was Found

#### **Short -Term**

If a solution is found to the road issue, the investors will get the opportunity to invest in it. In the scenario, if it is successfully implemented, the investors, as key contributors to the project, will gain a

good reputation in the short – term. Which can then be included in their monthly or annual reports and other publications like ESG reports. This has a positive short-term impact on investors.

### **Long-Term**

After the successful completion and implementation of the new improved road, the investing group will be recognized for their community work and will gain a positive reputation in the long term. As explained earlier under “Needs”, the same indirect profit pathway will apply for the investors and there is also a chance of the investment group getting additional incentives from the government in recognition of their community welfare involvement.

## **3.2) Customer and the Community**

### **3.2.1) Overview**

Customers and the community are one of the most important stakeholders. They have the highest interest compared to other stakeholders since they are the one benefiting the most from the project. They also have high power since they have the most influence in the project since they can protest or sign petitions if nothing is done. Customers and the community are the biggest supporters of this project since they are the ones most benefiting from it.

### **3.2.2) Examples of Customers and Community**

1. School children
2. Parents
3. Fishermen
4. Workers
5. Shopkeepers

### **3.2.3) Wants and Needs**

- One of the customers and community's wants is a proper functioning road which helps them get from place to place more easily.
- One of the needs of the customers and community is that the flooding problem during the wet season gets fixed since schools and shops will need to be closed and people would not be able to go outside to buy food or other necessary needs and it would also cause a lot of damage to building costing a lot of money to rebuild them and could also cause some deaths.

### 3.2.4) What Would Happen if a Solution Is Found

If there was a solution, then the customer and community would have a proper road which will make it easier for them to get from place to place and if the flooding problem is also fixed then schools and shops might not have to close down during the wet seasons and it would reduce the amount of buildings being destroyed and the impact of the damages done due to the flooding which would save them huge amounts of money.

### 3.2.5) What Would Happen if No Solution Is Found

If there were no solution found, then people would not have a proper functioning road and the floods would cause a lot of damage and money.

### 3.2.6) Concerns

One of the main concerns the customers and community have is how long it would take for the project to be finished since while the project is in Construction they might not have access to the road till it is completed. This would have an impact on them negatively in the short term.

## 3.3) Constructors and Suppliers

Builders/Constructors and Suppliers are two parties with a vested interest in developing local roadways. Their specific interests in the project are addressed below.

### 3.3.1) Constructors

Constructors are critical stakeholders who play an important role in executing engineering solutions and guaranteeing project success. They oversee carrying out plans developed by planners and engineers while taking into consideration the road's special conditions (*Holder, 2024*).

Constructors must be conscious of environmental implications when using their expertise and tools in road construction. Sustainable considerations must be prioritized to achieve responsible construction methods. Constructors have significant power in ensuring the project's success, but their interest in using the road may be lesser, given that they are members of the community.

#### 3.3.1.1) Concerns

One of the main concerns of constructors as stakeholders in this project is to carry out the development sticking to sustainable factors and minimizing ecological disruptions. Disturbances to the local community, such as noise and dust, should also be a concern. Weather conditions, the

development's surroundings, material availability, and receiving the materials on time are some other concerns of constructors as stakeholders.

#### *3.3.1.2) Needs*

Constructors need suitable guidelines to comprehend the development's aims, norms, and expectations. Relationships with the local community are crucial as they need support and cooperation while carrying out the project. They will prefer the best workers as they can conduct their work more efficiently. The skill level of workers is a higher concern. They would prefer the latest equipment and technology to overcome the challenges they will face due to the condition of the road and the island's climate.

#### *3.3.1.3) Wants*

Constructors will want their work to be environmentally friendly. They also prefer good relationships with locals and programs that will help them to be friendly with the locals. When they put their best effort into the success of the development, they will prefer timely payment and compensation in return. Recognition of the role of constructors in development can also be a desire of constructors as stakeholders.

#### *3.3.1.4) Effects on Constructors if a Solution is Found*

In the road development project for Saibai Island, constructors as stakeholders will have an effect if a solution for the situation of the roads is found. They can be short- and long-term and have positive and negative effects.

##### **Positive (short-term and long-term)**

If a solution is found and the project is conducted without delay, it will affect constructors. It will be cost-effective as the development can be done before the conditions of the road become worse. Overcoming challenges and developing the area properly will have a good reputation and it will affect their future opportunities. If the project satisfies the local community, constructors will have the support and appreciation of the locals. So, they will be able to carry out the project without any disturbances to finish it on time.

##### **Negative (short-term and long-term)**

It will be difficult for the constructors to explore new construction methods or technologies, as they are required to commence the project as soon as they can. There will be resistance from locals and

other stakeholders (a short-term effect). Initial costs for workers, equipment, and materials will be high, resulting in delays in the project commencement.

#### *3.3.1.5) Effects on Constructors if No Solution is Found*

If no solution is found for the muddy road condition on Saibai Island, it will also have short-term and long-term effects that will impact constructors as stakeholders positively and negatively.

##### **Positive (short-term and long-term)**

Being delayed is unacceptable but may have some positive short or long-term effects. Constructors will be able to adapt to their work surroundings and get familiar with the road's condition while maintaining good relationships with the locals. They will have time to explore more innovative development methods, such as new equipment and construction methods, as the world is getting more technologically advanced every second. Also, they will have time to learn about new approaches they can use. They can collectively devise various methods to overcome the challenges they will have to face while working on the project by analyzing the surroundings (Khoiry, n.d.).

##### **Negative (short-term and long-term)**

If no solution is found at the right time, the conditions of the road may be more severe than the current conditions. Then, the constructors may have to face new, more difficult challenges, leading to a negative impact on the construction company's reputation and financial losses. Delaying the project may be costly as more advanced methods, equipment, and additional labour must be put into the project to overcome the challenges. Conditions of the road and the surroundings being more severe may affect the safety of the constructors affecting the project development (Khoiry, n.d.).

#### **3.3.2) Suppliers**

Suppliers are also important stakeholders who play a major role in making this project successful. They oversee supplying the relevant goods for the development of the road. Quality materials should be supplied to obtain the best outcome for the project (Designing Buildings, n.d.). The main responsibility of the suppliers is to provide good-quality materials on time to continue the project without delay. Good quality materials are important for the durability and strength of the construction. If good quality materials are not provided, later they might impact the environment, affecting sustainability factors (Holder, 2024).

### *3.3.2.1) Concerns*

Suppliers have many concerns in a project like this. As they oversee supplying materials, they are concerned about the transportation challenges and costs for transportation as the island is in a remote location. They should ensure the quality of the materials and materials meet the necessary specifications for development. Like constructors, these people are also concerned about the environmental impact. Lastly, they are concerned about the fair and timely payment for their supplied materials and compensation for any unforeseen challenges. (Sharma, 2021)

### *3.3.2.2) Needs*

They need clear and proper guidelines and specifications to understand the quantity, quality, and types of materials that they should supply for the development. Friendly gestures from the local community and other stakeholders when they are supplying materials around the island.

### *3.3.2.3) Wants*

They should be assured that the materials provided are environmentally friendly. They are also concerned about the long-term opportunities in upcoming development projects and the recognition of the role of suppliers in contributing to the project's success.

### *3.3.2.4) Effects on Suppliers if a Solution is Found*

Improving the local access roads, as in the EWB challenge, needs lots of planning. If a solution is found by the planners and engineers, it will affect the material suppliers for this project. Those effects can be both positive and negative and those are discussed below.

#### **Positive (short-term and long-term)**

Suppliers can supply relevant materials for a demand when the project is ongoing. This will have a steady revenue stream. They become valuable partners and are recognized as an important part of the project. By adhering to sustainable practices their reputation will attract many other projects that are concerned about environmental conditions.

#### **Negative (short-term and long-term)**

They will have to adapt to quickly supplying the required materials on demand. Their initial investment in the inventories will be high. Also, they will need to adapt to the new and advanced materials to compete with their competitors.



### *3.3.2.5) Effects on Suppliers if No Solution is Found*

Suppliers will also have an effect if no solution is found, and those effects can also be positive and negative.

#### **Positive (short-term and long-term)**

Being delayed leads the suppliers to have the opportunity to collaborate with constructors and come up with materials that will be more cost-effective. If they have had some difficulties transporting materials without harming their quality, they will have time to come up with ideas to overcome those challenges. Also, they will have to come up with new business plans to expand their business through this project. They will also have time to manage their inventories to supply them on time when the project is ongoing.

#### **Negative (short-term and long-term)**

The uncertainty of the project will impact their revenue system and their financial stability. Their idea of expanding business by supplying quality materials and making contacts through the project will have a negative impact.

In conclusion, we can say that Suppliers and Constructors are two vital stakeholders that we should consider when we are implementing the project on improving local accessed roads in Saibai Island.

## **3.4) Local Government**

### *3.4.1) Overview*

Saibai Island is a locality within the Torres Strait Island Region local government area (Queensland, 2023). Hence, the Torres Strait Islands Regional Council (TSIRC), an elected local government body, is responsible for providing essential services to the Saibai Island. It plays a pivotal role in local governance and administration, overseeing the delivery of crucial services and programs within its area. The council's functions include a broad spectrum, including waste management, local infrastructure development, environmental management, and community development (TSIRC, n.d.-c). TSIRC collaborates with other stakeholders such as government agencies and community organizations, ensuring an inclusive decision-making process, aligning with community, making TSIRC a high-power stakeholder with low interest in this project.

The roles, functions, and responsibilities of TSIRC are established under the Local Government Act 2009. This legislative framework gives the council the authority to carry out its mandate and serve the local communities effectively. TSIRC aims for a future where the Torres Islands, including Saibai, thrive as resilient and sustainable communities (TSIRC, n.d.-b). TSIRC operates alongside local government bodies, such as the Torres Shire Council (TSC) and the Northern Peninsula Area Regional Council (NPARC). Additionally, the Torres Strait Regional Authority (TSRA) complements local governance efforts at a regional level, focusing on Torres Strait Islanders' economic, social, and cultural development (EWB Australia, 2024).

### 3.4.2) Concerns

The local government holds significant concerns regarding public safety, especially during the wet season when unsealed roads become hazardous. The risk of accidents due to muddy and impassable roads poses a notable threat to community safety. Additionally, there is apprehension about the impact of road conditions on education access, as flooding on school access roads can discourage students from attending classes, thereby affecting educational outcomes (EWB Australia, 2024). Infrastructure sustainability is also a focal point, with concerns about the longevity and integrity of the structures within the community

### 3.4.3) Interests

The local government expresses a strong interest in enhancing residents' overall quality of life. Improving road access aligns with this goal, ensuring community members can access essential services safely and easily. Furthermore, a vested interest in providing an environment conducive to education is evident, as improving road conditions directly supports better access to schools and contributes to students' educational success. Sustainable infrastructure development is also a key interest, focusing on projects exploring low-cost, locally sourced materials to create durable and cost-effective solutions for the community.

### 3.4.4) Needs

The local government requires essential resources and funding for infrastructure projects to address their concerns and fulfil their interests. Active community engagement is another critical need, as successful projects rely on collaboration with residents to understand their needs and involve them in decision-making processes. Given the structural assessment of the Holy Trinity Church, the local government may also require expertise and technical assistance in evaluating and addressing

infrastructure challenges, potentially through partnerships with engineering or construction experts (EWB Australia, 2024).

### 3.4.5) Goals

The primary goal of the local government is to establish safe and accessible infrastructure, particularly roadways, contributing significantly to community safety and ensuring residents can easily reach essential services, including schools. Another key objective is community engagement and satisfaction, involving residents in decision-making to align projects with local needs and preferences. The local government aims to implement long-term infrastructure solutions, exploring materials and construction methods that are effective in the short term, sustainable and easily maintainable within the local context (TSIRC, n.d.-c).

### 3.4.3) Impacts of the Project That Could Have on the Stakeholder

#### **If No Solution is Found**

In the short term, the absence of a solution would lead to continued disruptions in community movement, particularly during the wet season, as the unsealed and muddy roads pose safety risks. Students, in particular, face challenges attending school due to inaccessible roads, which impact their education poorly (EWB Australia, 2024). Safety concerns would increase, potentially resulting in accidents or injuries, thereby placing increased pressure on the Torres Strait Islands Regional Council (TSIRC) to address these immediate issues and respond to emergencies.

In the long term, persistent road accessibility issues could have a detrimental economic impact, hindering trade, transportation, and access to essential services, potentially leading to economic stagnation. Community frustration may escalate over time, eroding trust and support for the council among Saibai residents.

#### **If a Solution is Found**

With a successful solution, the short-term outlook improves significantly. Immediate improvements in road conditions lead to safer and more efficient community movement, particularly during adverse weather conditions. The local government's swift action in addressing road safety concerns demonstrates a commitment to the community's well-being, positively influencing its reputation in the short term. Furthermore, the positive educational impact is evident, as improved road access allows students to attend classes regularly, contributing to better educational outcomes.

In the long term, sustainable road solutions contribute to community development by fostering economic growth and improving living standards. The success in resolving the road access challenges builds trust and support for the Torres Strait Islands Regional Council among community members, creating a positive and enduring relationship that benefits Saibai Island over the long term.

## **4.0) Design Requirements**

### **4.1) Design Criteria and Constraints**

#### **4.1.1) Sustainability**

##### *4.1.1.1) Criteria*

For our road project, sustainability is a major concern that must be addressed for a complete practical implementation of it. This is a criterion that is most sought by many of the stakeholders in the project. For example, investors who fund the project want the road to be as sustainable as possible since its sustainability affects their reports.

In our scenario, sustainability can be broken down into 2 main concepts, they are environmental and social aspects. Under the environmental aspect, attention should be given to the use of natural resources in a manner that does not harm the natural habitats and ecosystems. The materials used in the design should be sourced and used with the greatest attention to minimal environmental pollution. Thus, it highlights the importance of selecting sustainable materials that are low cost and less harmful to the environment.

When considering the social aspect of sustainability, the local community should have the ability to easily maintain the road with minimum cost and natural resources that can be found on the island itself. For the practical implementation of this project, the highest priority should be given to the customer's need, which in our case is the local community that uses the road. The safety of the community should be assured in both construction time periods and afterwards. Since this is a road project, the construction phase will require a considerable amount of time; thus, during this period, the negative effects on the locals as well as the environment should be minimized as much as possible.

#### *4.1.1.2) Constraint*

The main constraint in our project regarding sustainability is sourcing materials with the least pollution to the environment and that also be durable enough. Sourcing a small amount of raw materials would not be a problem since the environmental damage done will be negligible. However, when we want to source raw materials needed for large-scale construction, pollution will mostly be inevitable. So as a result, more focus should be given to that aspect.

#### **4.1.2) Safety**

As per the needs of the key stakeholders to improve access to muddy roads for the community in Saibai Island, especially including students, safety enhancement in the project is crucial. Under safety enhancement, 4 main criteria and 2 constraints will be considered.

##### *4.1.2.1) Criteria*

##### Flood resistance

Flood resistance is a primary need in ensuring safe road access, particularly during heavy rainfall and storm surges on Saibai Island. The design must incorporate proper drainage systems, raised road surfaces, and flood-resistant materials to prevent road inundation and maintain accessibility during inclement weather conditions. The design should implement effective flood resistance measures, to minimize the risk of road closures and ensure continuous access for the community, including students attending schools.

##### Skid resistance

Skid resistance is essential to reduce the risk of accidents, especially on muddy roads prone to slippery conditions. The road surface must provide adequate traction to prevent vehicles from slipping, particularly during wet weather. Textured surfaces or suitable additives should be incorporated into the road construction to improve traction and enhance skid resistance. With proper skid resistance implementation, Proper skid resistance implementation enhances safety and prevents enhance safety and prevent road accidents, contributing to improved road accessibility for the community.

##### Structural stability

Structural stability is crucial for ensuring the long-term safety and durability of road infrastructure. The design must prioritize structural integrity, considering factors such as heavy traffic loads, erosion, and environmental conditions. Proper compaction of materials, reinforced structures, and regular

maintenance are essential to ensure structural stability and resilience. Minimizing the risk of road failures and ensuring safe passage for road users, including pedestrians and vehicles, can be achieved by prioritizing structural stability.

### Pedestrian safety

Pedestrian safety is a key consideration, particularly near school zones, to ensure the safety of students and other pedestrians. The design must incorporate safe pedestrian pathways, crossings, and designated areas to separate pedestrians from vehicular traffic. To facilitate safe pedestrian movement, sidewalks, pedestrian bridges, or marked crossings should be implemented.

#### *4.1.2.2) Constraints*

The project must adhere to safety regulations, remain under budget, and use reasonably priced, locally produced components. Additionally, it must employ efficient waste management and conservation techniques, use sustainable materials, and reduce its ecological impact on Saibai Island.

### **4.1.3) Materials**

#### *4.1.3.1) Criteria*

Materials are one of the most important things to consider when making this project. Material will need to be cheap enough, durable enough to hold tons of heavy weight and will need to sustain rain and flooding conditions and not wear or tear. The main stakeholders that will need to consider these are the ones funding it such as the local government and the investors.

Most of the materials will need to be cheap enough to fit our budget so most of the materials should be found on the island itself or from somewhere near the island otherwise it would cost a lot of money to import such materials to the island.

#### *4.1.3.2) Constraints*

Some of the constraints in our project for materials are that we need to source our materials on the island as much as we can. We should be careful when taking certain materials in bulk from the island so that we do not drain the island from its natural resources (such as deforestation) so that there would still be these materials available for future use for the islanders. These materials will also need to be sustainable and durable.

#### **4.1.4) Durability**

##### *4.1.4.1) Criteria*

Durability is an important design consideration for our project to improve local access roads on Saibai Island, especially during the wet season. The chosen stakeholders, including investors and local government, particularly TSIRC, place a high value on design durability, preferring a long-lasting solution that can deal with a variety of elements that cause wear and tear. Generally, these include sun exposure, strong rainfall, and frequent traffic. As a result, planners must examine effective strategies to handle these difficulties while also ensuring the project's sustainability. Efforts are being undertaken to ensure the project's success and long-term relevance. To attain this purpose, several critical elements must be considered. First and foremost, constructors and planners must properly comprehend the climate and environmental circumstances in which the project will be implemented. Furthermore, selecting appropriate materials for different sections of the road is critical. For example, utilizing riprap and sandbags as a base can make the road more durable. Furthermore, installing a strong drainage system is critical to preventing water collection on the road, especially given the heavy rains in the area where the project is being carried out. Considering these concerns and using appropriate techniques, the project may ensure the road's durability while meeting stakeholders' expectations for long-term effectiveness.

##### *4.1.4.2) Constraints*

Some of these limits are vegetation management and slope stability along the route. Heavy rains and flooding can readily destroy vegetation-covered slopes, making slope stability difficult to maintain. As a result, while designing road construction projects for long-term use by the local population, it is critical to explore solutions to these concerns.



## 4.2) Design Criteria Weighting Summary Table

The following table shows the design criteria, their weighting and the attributes considered for each of the design criteria.

Table 4: Design criteria and the weighting

| Criteria       | Weight | Attributes considered                                                                                                                                                                                                                                 |
|----------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sustainability | 20%    | <ul style="list-style-type: none"><li>○ Use of low carbon material (5%)</li><li>○ Low maintenance leading to less material usage for repair and replacement (5%)</li><li>○ User friendliness (5%)</li><li>○ Enhanced social acceptance (5%)</li></ul> |
| Safety         | 30%    | <ul style="list-style-type: none"><li>○ Use of robust materials (7%)</li><li>○ Provision of access to neighborhood (7%)</li><li>○ Follow road safety (8%)</li><li>○ Considerations for pedestrians' safety (8%)</li></ul>                             |
| Durability     | 30%    | <ul style="list-style-type: none"><li>○ Design considerations (7%)</li><li>○ Use of durable material (7%)</li><li>○ Provision of erosion control (8%)</li><li>○ Considerations for flood control (8%)</li></ul>                                       |
| Materials      | 20%    | <ul style="list-style-type: none"><li>○ Use of local material (6%)</li><li>○ Recycled content in material (6%)</li><li>○ Use of durable material (8%)</li></ul>                                                                                       |

## 5.0) Design Concept Solution

### 5.1) Concept Outline

#### 5.1.1) Scoring justification

Four main criteria namely sustainability, safety, durability and materials were considered in designing the concept design. In order to ensure long-term viability and minimize environmental effects, sustainability is essential. Safety considers the needs of road users first, improving useability and accessibility. In difficult circumstances, durability ensures the resilience of the infrastructure. The choice of materials encourages economical use and the preservation of local resources.

Table 5 and 6 demonstrate how each individual concept design was evaluated using the design criteria. All the individual concept drawings are included in appendix B.

*Table 5: Rovindu Tharin's (Conceptual Design 1) and Yameen Shariz's (Conceptual Design 2) Conceptual Designs*

| Criteria       | Weight      | Attainable weight according to the attributes considered |                     |
|----------------|-------------|----------------------------------------------------------|---------------------|
|                |             | Conceptual Design 1                                      | Conceptual Design 2 |
| Sustainability | 20%         | 16%                                                      | 12%                 |
| Safety         | 30%         | 22%                                                      | 21%                 |
| Durability     | 30%         | 12%                                                      | 23%                 |
| Materials      | 20%         | 18%                                                      | 17%                 |
| <b>Total</b>   | <b>100%</b> | <b>68%</b>                                               | <b>73%</b>          |

*Table 6: Hashin Maharambage's (Conceptual Design 3) and Chiranga Karunaratne's (Conceptual Design 4) Conceptual Designs*

| Criteria       | Weight      | Attainable weight according to the attributes considered |                     |
|----------------|-------------|----------------------------------------------------------|---------------------|
|                |             | Conceptual Design 3                                      | Conceptual Design 4 |
| Sustainability | 20%         | 14%                                                      | 12%                 |
| Safety         | 30%         | 24%                                                      | 26%                 |
| Durability     | 30%         | 25%                                                      | 28%                 |
| Materials      | 20%         | 15%                                                      | 13%                 |
| <b>Total</b>   | <b>100%</b> | <b>78%</b>                                               | <b>79%</b>          |

Table 7 reveals the attainable weight out of the allocated weight for each design criteria of the finalized concept design.

Table 7: Design Criteria and Weighting for the Final Design

| Criteria       | Weight      | Attainable weight |
|----------------|-------------|-------------------|
| Sustainability | 20%         | 15%               |
| Safety         | 30%         | 27%               |
| Durability     | 30%         | 28%               |
| Materials      | 20%         | 16%               |
| <b>Total</b>   | <b>100%</b> | <b>86%</b>        |

### 5.1.1) Structure of the Road

When building a road, the main steps of the process would follow as; digging and excavation for removal of soil and rocks and levelling, construction of embankments, tunnels, and bridges as necessary and the pavement structure is placed utilizing a variety of different pieces of construction equipment and material. Essentially, the purpose of the pavement structure is to support the load vehicles and pedestrians and to distribute it to the roadbed. So, it is necessary to produce a pavement structure that is very well-bonded, and for this purpose, the road pavement structure consists of a sub-grade, sub-base, base course, binder course and surface course going from bottom to top layer, respectively (*Construction Cost, n.d.*).

Purpose of each course;

1. Sub-grade- the finished earthwork that has been compacted. It provides support for the pavement. All pavement loads, including traffic, are carried by the sub-grade.
2. Sub-base- A layer of granular material between the sub-grade and the base course. It improves the bearing capacity of the sub-grade, aids in drainage, and helps prevent the capillary rise of subsoil water.
3. Base course- This layer lies immediately beneath the surface or binder course and is typically made of crushed aggregate or HMA (Hot Mix Asphalt). It's designed to withstand high shearing stresses developed due to traffic impact and distribute loads across a wider area.

4. Binder course- Also known as the base coat or intermediate course, this layer of hard stones is provided between the base course and the wearing course. It transmits the load over a large area of the base course and acts as a layer of transmission material.
5. Surface course- The topmost layer directly exposed to traffic, also known as the wearing course. It provides a smooth, durable surface that is impervious to water, ensuring that water does not enter the base course. It needs to be resilient enough to withstand heavy loads and abrasive forces.

It was decided on the following material and depth of layers in the proposed design solution, considering the vehicle density, light weight vehicles usage, materials combinations used and the cost-effective alternatives (*Construction Cost, n.d.*).

*Table 8: Description of the layers of the road design (Construction Cost, n.d.)*

| <b>Pavement layer</b> | <b>Selected material</b>   | <b>Depth of the layer(cm)</b> |
|-----------------------|----------------------------|-------------------------------|
| Sub-base              | Riprap                     | 50                            |
| Base course           | Stabilized soil            | 40                            |
| Binder course         | Reclaimed asphalt pavement | 20                            |
| Surface course        | Micro-surfacing            | 1.3                           |

With each layer's chosen material and depth, the road is proposed to be elevated 1.1 meters from the ground level. The elevation of the road, together with the drainage system (discussed in section 5.1.4) acts as a prevention of the floods caused by rain and tidal waves.

### **Material details**

#### *Riprap*

Sub-base layer of 500 mm, comprising of riprap material uses large, angular stone, cobbles, or boulders to provides a strong and stable foundation for the road, enhancing its durability and resistance to deformation under heavy load bearings (*Granite Co., n.d.*).

#### *Stabilized soil*

Base course layer of 40mm, made up of stabilized soil improves soil quality for road construction by enhancing its properties and ensuring suitability. The proposed Stabilized soil content is locally available soil mixed with 15% cement kiln dust stabilizer. This mixture type was chosen as it has a low

energy consumption rate of 16.3 MJ/Ton and a low equivalent CO<sub>2</sub> emission of 2.6 Kg/Ton, contributing to the design criteria of sustainability as well as materials (*Ghavami et al., 2020*). Stabilized soil, forming the base course, distributes loads effectively, boosting road strength and durability. Additionally, it prevents soil movement, maintains subgrade stability, and ensures a secure road surface (*PCC Pavement, n.d.*).

#### *Reclaimed Asphalt Pavement (RAP)*

The binder course layer of 200mm, incorporates reclaimed Asphalt Pavement (RAP) is a great binder course for many reasons. Firstly, RAP offers sustainability by utilizing existing asphalt during reconstruction and resurfacing, promoting the eco-friendly use of materials. Secondly, it proves cost-effective, reducing project expenses while maintaining environmental consciousness. Moreover, RAP demonstrates high performance, often matching or surpassing the quality of virgin mixes when properly planned and executed. Additionally, RAP enhances the strength and stiffness of pavement bases and sub-bases, mitigating failures such as rutting and fatigue cracking induced by dynamic loads (*Mariyappan et al., 2023*).

#### *Micro-surfacing*

Micro-surfacing as a surface course of 13mm, offers several benefits to pavement maintenance and construction projects. It extends the lifespan of asphalt pavements by enhancing skid resistance, preventing moisture intrusion, and shielding against oxidation and raveling. Additionally, it restores the uniform black appearance of roads, enhancing aesthetics. Micro-surfacing is the most economical option for levelling needs. It boasts quick construction times with minimal disruption to the travelling public, making it a preferred choice for efficient pavement management (*Road Resource, n.d.*).

On the finished surface course, it is designed to have two white road marking lines by the sides of the road (*figure 10- view 05*). Road markings play a crucial role in directing and managing traffic, acting as a psychological barrier, and designating the boundaries of the traffic flow. Additionally, it directs pedestrian traffic into designated areas (*TrafficInfraTech - Editor, 2021*).

### **5.1.2) Embankment**

The proposed solution introduces road embankments, which provide structural support and maintain the desired level of the roadway above the surrounding ground. They also aid in flood control by preventing water from overflowing onto the roadway (*GceLab, n.d.*). In the design solution, the

embankments are provided with a 1:2(V:H) side slope for the stability of the embankment (PRS Geo-Technologies, 2022).

### **Erosion control**

The embankment is at elevated risk of erosion without any erosion control measures. The concentrated forces produced by the flowing water during a flood can lead to the formation of holes, longitudinal and transversal furrows and undercutting at the base. Then, the soil may slip down onto the canals, subsequently affecting the water flow of the culvert (refer to *section 5.1.4*). Surface vegetation is absolutely the most environmentally friendly solution, but it is vulnerable and easily erodible. Therefore, colonization by vegetation is necessary but rather difficult.

The design solution suggests geomat as the best erosion control strategy. Geomat comprises three layers of extruded polypropylene grids that are subsequently bi-oriented. The three layers of geomat are then sewed together, overlapped, and tied off using black polypropylene yarn. The lower portion of the geomat might be used to build the biodegradable pre-seeded cloth. To keep warm for the growth of seedlings inside the cloth or to create an integrated green landscape, it is either black or green in colour. As a result, the geomat first creates an efficient slope protection layer, quickly covers the grass, and effectively anchors the roots to the soil and the geomat. (Anping County Wennian Wire Mesh Products Co., n.d.).

#### **5.1.3) Cross-fall**

In addition to the embankments, road cross-fall is necessary for road pavement design to ensure adequate surface water removal from the road to the roadside drains. Therefore, building the road at a cross-slope of 6% is proposed, which means 0.72" of fall per horizontal 12" of road width measured from the centerline toward the roadside drains (New Zealand Forest Owners Association, n.d.).

#### **5.1.4) Drainage System**

Drainage systems are essential for roads to function properly for their service life. In flowing water, pavement flaws such as subgrade erosion may arise. Therefore, water should not collect on top of or beneath the pavement. Water must be able to effectively flow off of the road with the right cross profile. To prevent the subgrade from being eroded, there should be drains on both sides of the sidewalk. Water should also be moved and disposed of appropriately through a drain. (Housing, 2023).

Roadside drains are installed to convey rainwater from the road surface and roadbed to an outlet. Therefore, in the proposed solution, roadside drains with gratings are installed for water runoff and the users' safety. These roadside drains will direct the water runoff to the main water outlet of the drainage system; the culvert drains through the side drains (*figure 10- view 05*). The side drains will repeat every 100 m on both sides of the road.

As the main flood control and tidal wave control measure, a culvert is proposed in the solution. In addition to these functions at low tides, the culvert can function as a wildlife underpass. The culvert drain is used to direct the water runoff from the roadside drains and the water from the culvert to dispose of all the collected water into the lagoons or the sewer access hole, depending on the closest option.

Determining the optimal spacing between road culverts was based on the data obtained on the rainfall and the tidal waves. It was assumed that the region experiences fast lateral flows. For permanent roads, the recommended maximum culvert spacing is 100 m, ensuring efficient water diversion and minimizing erosion risks, hence the proposed design will have a spacing of 100 m between two culverts. Culverts should have a diameter of at least 800 mm in the chosen water flow, hence the design solution proposes culverts with a diameter of 1 meter (*Adequate Cross Drainage and Subsurface Flow Capacity – Green Roads for Water, n.d.*).

#### **5.1.5) Pedestrian walkways**

When building a pedestrian walkway, factors such as the safety of the pedestrians, site conditions, and pavement durability should be considered. Safety is a crucial criterion in making pedestrian walkways. Pedestrian walkways must be well-designed so that pedestrians and motorists can recognize the relevant paths to reduce the chance of accidents. The designed solution is mainly based on these criteria. As the road is elevated from the normal ground level, it was decided to build a green wall (*further explained under section 5.1.6*) beside the pedestrian walkway, near the slope as a safety measure to prevent the risk of pedestrians falling. Between the pedestrian walkway and motorist pathway, there is a grating over the drain, which helps identify the two pathways clearly. Additionally, the white road marking lines play an important role in pedestrian safety by acting as a visible barrier between vehicles and pedestrians.

On the embankments, there are stairways where pedestrians can access the elevated road. This stairway is connected to a crossing bridge over the culvert drain (*figure 10-view 06 and figure 9-view*

04). The crossing bridge is equipped with a safety fence, made from bamboo. These stairways will repeat every 100 m on both sides.

For the construction of the pedestrian walkway, road access stairway and crossing bridge, it is proposed to be built with concrete. Unlike other materials, concrete can withstand heavy foot traffic without significant wear and tear, making it ideal for high-traffic pedestrian areas. In addition to its durability, concrete has other advantages, such as a slip-resistant surface preventing pedestrians from slipping, minimal maintenance once installed, and versatility of the material, making concrete the ideal pavement material for the proposed design solution. With a well-designed and built pedestrian walkway, the community on Saibai Island is able to have road access in any weather condition.

#### **5.1.6) An Innovative Idea: A green Wall with Unique Combination of Hydrogel and Bamboo**

As for the innovative idea, it is proposed to build a green wall, which serves the following objectives;

- o Further erosion control- Enhancing erosion control through the integration of live bamboo roots, renowned for its unique traits and soil-stabilizing properties.
- o Air purification- Plants absorb CO<sub>2</sub>, carbon monoxide, and toxins through photosynthesis, lowering CO<sub>2</sub> levels in urban areas with ample vegetation.
- o Dust suppression- Greenery suppresses dust, enhancing air quality near construction sites and highways, potentially reducing respiratory illness. London's recent efforts to meet EU (European Union) regulations include a focus on reducing PM<sub>10</sub> particles.
- o Heat reduction- Increasing vegetation in urban hot spots, like urban canyons, cools trapped air and reduces reflected heat, with temperature differences of up to 17°C observed between hard and vegetated surfaces in the same location (Biotecture, 2023).

The green wall is built using both treated bamboo and live bamboo as the structure with the treated bamboo standing on a concrete base for sturdiness. The bamboo is set to be approximately 1.2 m apart, and coir ropes are woven on the bamboo, providing support for the plants to grow on (*figure 9- view 03*). Using live bamboo helps prevent soil erosion and maintain soil stability with its soil-grabbing roots. Also, its roots could leach heavy metals from the soil, and efficiently draw water closer to the surface due to its strong water absorption capability (Tardío et al., 2018). This soil drawing



ability acts as a great support for the plants on the green wall. A mix of live and treated bamboo reduces the water required for the green wall.

Additionally, for every 10 m, in the place of bamboo, it is designed to have a solar street lighting unit. Street lighting enhances safety and security for both pedestrians and motorists. The advantages of using solar street lighting are that they are reliable, do not require any external power source and easy to install, do not require any complicated wiring or trenching (*An In-depth Article on the Pros and Cons of Solar Street Lighting Can Help You Make the Best Choice*, n.d.).

A simple irrigation system, consisting of an irrigation pipe will supply the water requirement of the green wall, and the irrigation system's source will be from the island's water supply (*figure 9- view 03*). To optimize the water supply to the green wall, hydrogel crystals are proposed to mix with the soil for the green wall. Hydrogels are a water-absorbing polymer that absorbs and retains water in as much as 80-180 times the original volume, only to release the water to plant roots at a later time (Bowman et al., 1991).

*Hedera helix* (English Ivy) and Virginia Creeper (*Parthenocissus quinquefolia*) are chosen in the proposed green wall design. These plants serve the objective of air purification as they absorb more CO<sub>2</sub> than regular plants and remove volatile organic compounds (VOCs) and particulate matter through their leaves. Additionally, their dense foliage provides surface area for capturing airborne particles, effectively filtering the air. These plants contribute to improved air quality, and they are hardy, fast-growing, and adaptable to various conditions, making them suitable for roadside environments. Additionally, they provide lush green foliage and can help soften structures' appearance while offering wildlife habitat (The Joy of Plants, n.d.).

The proposed green wall further prevents erosion with live bamboo and optimizes the water demand of the green wall with hydrogel crystals, highlighting a combination of existing concepts to utilize a unique addition to the community of Saibai Island.

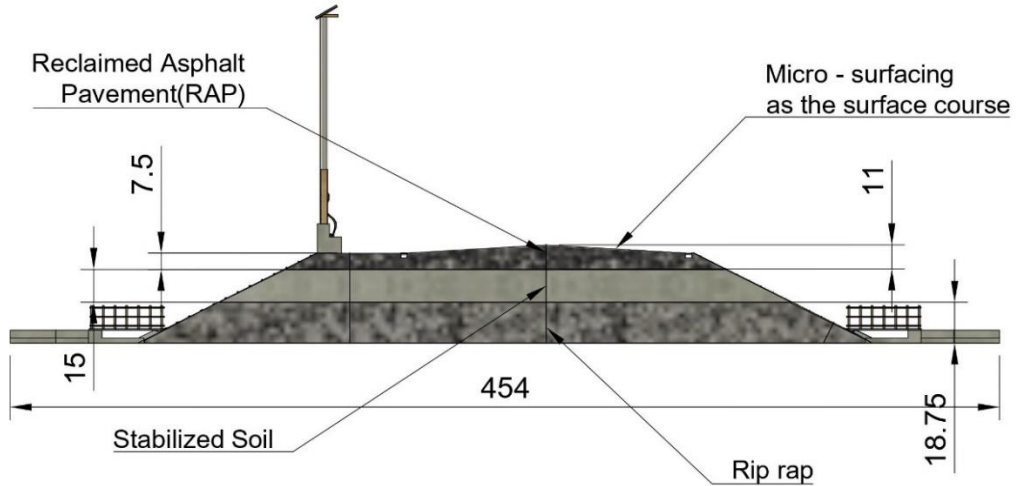
### **5.3) Concept Drawings**

This section will offer a visualization of our proposed design, supported by descriptive drawings that highlight the main ideas and features of the design. These illustrations are meant to improve comprehension and give a clear picture of how each part works together.

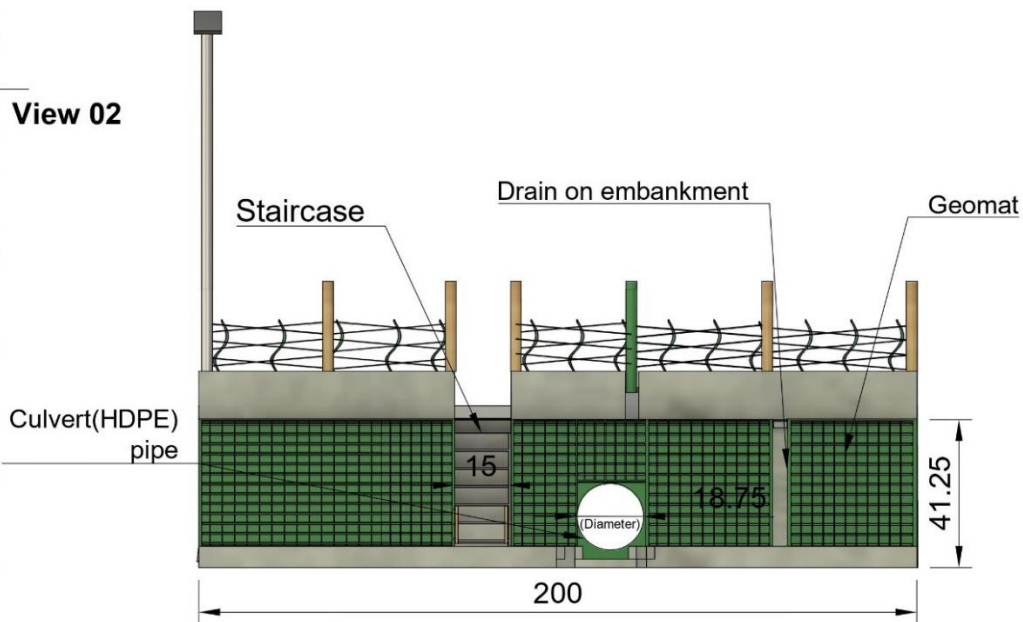
# Concept Drawing

## Drawing 01

### View 01 - Layers of the road



### View 02



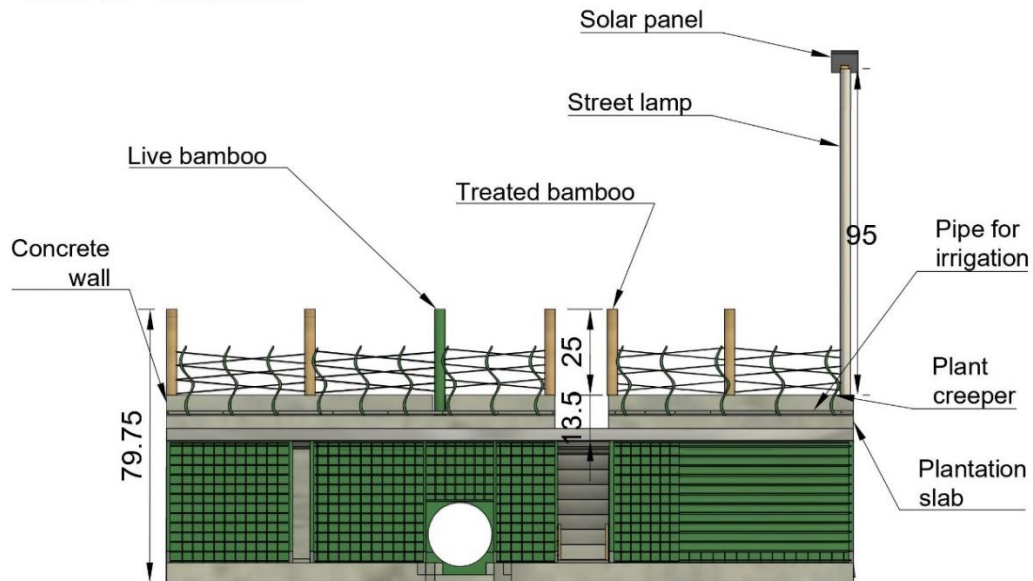
|                                                                     |                     |                                   |                 |
|---------------------------------------------------------------------|---------------------|-----------------------------------|-----------------|
| Dept.                                                               | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by     |
| <b>Scale</b><br><b>1 mm : 27 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status |
|                                                                     |                     | Title<br><b>INDE Road Design</b>  | DWG No.         |
|                                                                     |                     | Rev.                              | Date of issue   |
|                                                                     |                     | Sheet                             | <b>D1</b>       |

Figure 6: Concept drawing- layers of the road and side view

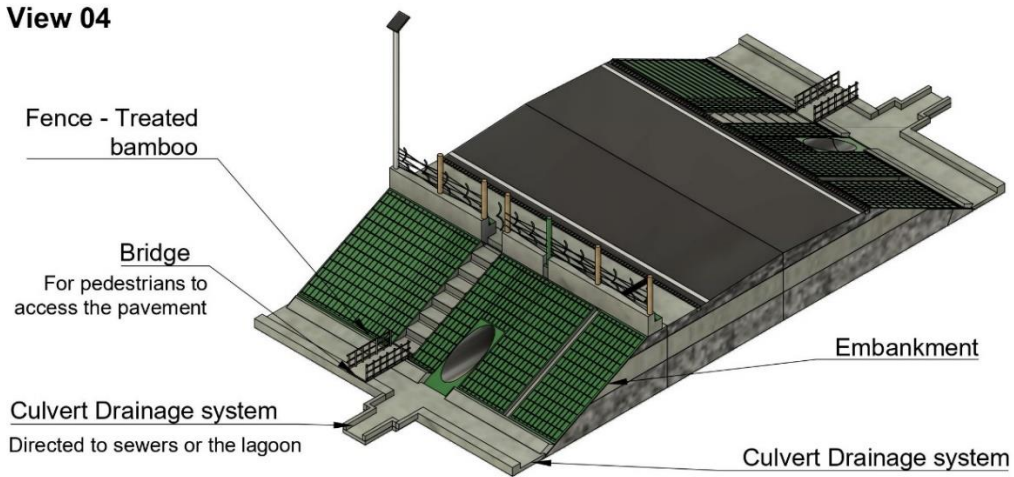
# Concept Drawing

## Drawing 02

### View 03 - Green wall



### View 04



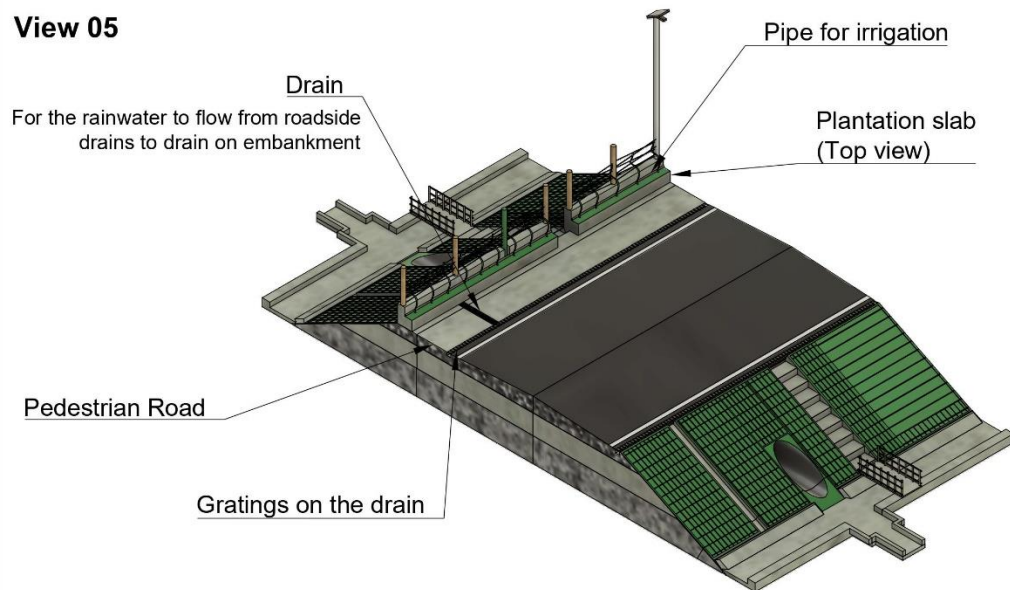
|                                                                     |                     |                                   |                 |
|---------------------------------------------------------------------|---------------------|-----------------------------------|-----------------|
| Dept.                                                               | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by     |
| <b>Scale</b><br><b>1 mm : 27 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status |
|                                                                     |                     | Title<br><b>INDE Road Design</b>  | DWG No.         |
|                                                                     |                     | Rev.                              | Date of issue   |
|                                                                     |                     | Sheet                             | <b>D2</b>       |

Figure 7: Concept drawing- green wall and orthogonal view

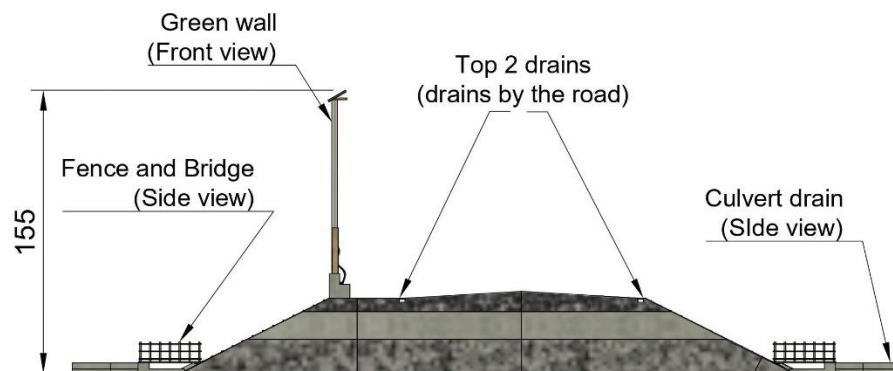
# Concept Drawing

## Drawing 03

### View 05



### View 06



|                                                                     |                     |                                   |                 |
|---------------------------------------------------------------------|---------------------|-----------------------------------|-----------------|
| Dept.                                                               | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by     |
| <b>Scale</b><br><b>1 mm : 27 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status |
|                                                                     |                     | Title<br><b>INDE Road Design</b>  | DWG No.         |
|                                                                     |                     | Rev.                              | Date of issue   |
|                                                                     |                     | Sheet<br><b>D3</b>                |                 |

Figure 8: Concept drawing- orthogonal view and corresponding drainage system

## 5.2) Concept Budget

Table 9 shows the detailed budget for the materials required in the design for a 100 m road stretch. All the prices are given in Australian dollars and given for a sample of the materials that should be used in the design.

Table 9: Concept budget for a 100 m road stretch (without labor cost)

| Material                         | Use                                                                      | Reference                                                                                                                                                                                         | 01 Unit                                                                                | Unit Price (AU\$) | Quantity                                                      | Amount (AU\$) |
|----------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------|---------------|
| Cement bags                      | For the construction of drains, stairway, bridge and pedestrian walkaway | <a href="https://bricksblockspaversonline.com.au/products/cement-australia-gp-cement-20kg-bag-p233">https://bricksblockspaversonline.com.au/products/cement-australia-gp-cement-20kg-bag-p233</a> | 1 bag (20kg bag)                                                                       | 9.93              | 2358 bags<br>(refer Appendix B for the calculation)           | 23,414.94     |
| Fill Sand                        | For the use of drains, stairway, bridge and pedestrian walkaway          | <a href="https://www.turtlenursery.com.au/sand-supplies/fill-sand/">https://www.turtlenursery.com.au/sand-supplies/fill-sand/</a>                                                                 | Per cubic meter                                                                        | 88.11             | 235.76 cubic meters<br>(refer Appendix B for the calculation) | 20,772.81     |
| 20mm Aggregates                  | For the construction of drains, stairway, bridge and pedestrian walkaway | <a href="https://mjrowles.com.au/products/aggregate">https://mjrowles.com.au/products/aggregate</a>                                                                                               | Per cubic meter<br>(The density of aggregate was considered as 1500kg/m <sup>3</sup> ) | 99.90             | 235.76 cubic meters<br>(refer Appendix B for the calculation) | 23,552.42     |
| Construction Concrete Reinforced | For the construction of drains, stairway, bridge and                     | <a href="https://www.alibaba.com/product-detail/1600758993331.h">https://www.alibaba.com/product-detail/1600758993331.h</a>                                                                       | Per Ton                                                                                | 69.82             | 7.915 Ton<br>(refer Appendix B)                               | 552.63        |

|                                        |                                                    |                                                                                                                                                                                                                                                                                                                 |                                     |                                                         |                      |        |
|----------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------|----------------------|--------|
| Deformed Steel Rebar                   | pedestrian walkaway                                | <a href="https://www.alibaba.com/product-detail/Wholesale-Cheap-Price-Natural-Bamboo-Stakes_50035218513.html?spm=a2700.7724857.0.0.66847fcciSs3mk">tml?spm=a2700.7724857.0.0.66847fcciSs3mk</a>                                                                                                                 |                                     |                                                         | for the calculation) |        |
| Treated Bamboo                         | For the green wall                                 | <a href="https://www.alibaba.com/product-detail/Wholesale-Cheap-Price-Natural-Bamboo-Stakes_50035218513.html?spm=a2700.7724857.0.0.66847fcciSs3mk">https://www.alibaba.com/product-detail/Wholesale-Cheap-Price-Natural-Bamboo-Stakes_50035218513.html?spm=a2700.7724857.0.0.66847fcciSs3mk</a>                 | Per pole                            | 0.50                                                    | 200 poles            | 100.00 |
| Biodegradable coir ropes               | For the mesh of green wall                         | <a href="https://www.abundancefarm.com.au/products/biodegradable-coir-rope">https://www.abundancefarm.com.au/products/biodegradable-coir-rope</a>                                                                                                                                                               | 5 mm x 20 m                         | 3.99                                                    | 300 m                | 59.85  |
| Micro-surfacing                        | For the surface of the topmost layer of the road   | <a href="https://savemyroad.com/treatment-resource-center/premium-micro-surfacing/">https://savemyroad.com/treatment-resource-center/premium-micro-surfacing/</a>                                                                                                                                               | Per square meter                    | 6.00                                                    | 9.31 square meters   | 55.90  |
| Stabilized soil-Cement Kiln Dust (CKD) | For the 2nd from the bottom-most layer of the road | <a href="https://www.cementaustralia.com.au/">https://www.cementaustralia.com.au/</a>                                                                                                                                                                                                                           | -                                   | Free of charge (as CKD is wastage from cement industry) | 37.5 cubic meters    | -      |
| HDPE pipes                             | For the culverts                                   | <a href="https://www.alibaba.com/product-detail/PE-Double-Wall-Corrugated-Road-Culvert_1600845606873.html?spm=a2700.details.you_may_like.2.182a5c42IFyfZ1">https://www.alibaba.com/product-detail/PE-Double-Wall-Corrugated-Road-Culvert_1600845606873.html?spm=a2700.details.you_may_like.2.182a5c42IFyfZ1</a> | 1 Meter (diameter) x 8.5 m (length) | 34.55                                                   | 1 unit               | 34.55  |

|                                          |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                      |        |                   |           |
|------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|-------------------|-----------|
| Heavy-Duty Stainless-Steel Floor Grating | To cover the gutters on the sides of the road | <a href="https://www.alibaba.com/product-detail/Heavy-Duty-Steel-Floor-Grating-Anping_60731960130.html?spm=a2700.galleryofferlist.p_offer.d_title.42ab51375Ncly5&amp;s=p">https://www.alibaba.com/product-detail/Heavy-Duty-Steel-Floor-Grating-Anping_60731960130.html?spm=a2700.galleryofferlist.p_offer.d_title.42ab51375Ncly5&amp;s=p</a>                                                                             | Per piece (700 cm x 9.5 cm)                                                          | 13.87  | 30 pieces         | 416.10    |
| Recycled Asphalt (RAP)                   | For the topmost layer of the road             | <a href="https://homeguide.com/posts/recycled-asphalt-millings-cost">https://homeguide.com/posts/recycled-asphalt-millings-cost</a>                                                                                                                                                                                                                                                                                       | Per cubic meter<br><br>(The density of RAP was considered as 1800kg/m <sup>3</sup> ) | 40.70  | 100 cubic meters  | 4,070.00  |
| Rip rap                                  | For the bottom-most layer of the road         | <a href="https://nclsupply.com/product/rip-rap-per-ton-cubic-yard/">https://nclsupply.com/product/rip-rap-per-ton-cubic-yard/</a>                                                                                                                                                                                                                                                                                         | Per Cubic meter                                                                      | 53.06  | 405 cubic meters  | 21,489.30 |
| Erosion mat                              | For the slope to prevent erosion              | <a href="https://sdlxgeosynthetics.en.made-in-china.com/product/BOmfNSbYEDrM/China-3D-Erosion-Control-Mat-for-Prevent-Soil-Erosion.html">https://sdlxgeosynthetics.en.made-in-china.com/product/BOmfNSbYEDrM/China-3D-Erosion-Control-Mat-for-Prevent-Soil-Erosion.html</a>                                                                                                                                               | Per square meter                                                                     | 1.23   | 212 square meters | 260.76    |
| Solar powered streetlight                | Streetlighting                                | <a href="https://ydelectric.en.made-in-china.com/product/VnSpDtwbAOhP/China-Premium-Street-Light-Poles-for-Outdoor-Lighting-Solutions.html?ads_tp=dingtui&amp;srv_id=vQLmgbTrBxnc&amp;delivery_region=5&amp;pv">https://ydelectric.en.made-in-china.com/product/VnSpDtwbAOhP/China-Premium-Street-Light-Poles-for-Outdoor-Lighting-Solutions.html?ads_tp=dingtui&amp;srv_id=vQLmgbTrBxnc&amp;delivery_region=5&amp;pv</a> | Per streetlight                                                                      | 154.80 | 9                 | 1,393.20  |



|                                                                                                                                                                                                                                                                                  |  |                                                          |  |  |  |                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------|--|--|--|----------------------------------|
|                                                                                                                                                                                                                                                                                  |  | <a href="#">_id=1hr9s32nc7f3&amp;faw_id=1hr9s74phf1d</a> |  |  |  |                                  |
| <b>Total (AU\$)</b>                                                                                                                                                                                                                                                              |  |                                                          |  |  |  | 96,172.46                        |
| <b>Maintenance cost (every 3 years) (AU\$)</b><br>(Average Annual Cost for Road Maintenance by Operational Maintenance Level-<br><a href="https://www.fs.usda.gov/Internet/FSE_DOCUMENTS/fseprd528063.pdf">https://www.fs.usda.gov/Internet/FSE_DOCUMENTS/fseprd528063.pdf</a> ) |  |                                                          |  |  |  | 3,553.92<br>(Annual cost per km) |

## 6.0) Model Drawing and Specification

This section will discuss the design and documentation of a scaled-down representative model to effectively convey the core features and innovative aspects of the proposed solution. This model will serve as a crucial proof of the concept, facilitating a clear demonstration of the principal components and innovations incorporated in the design.

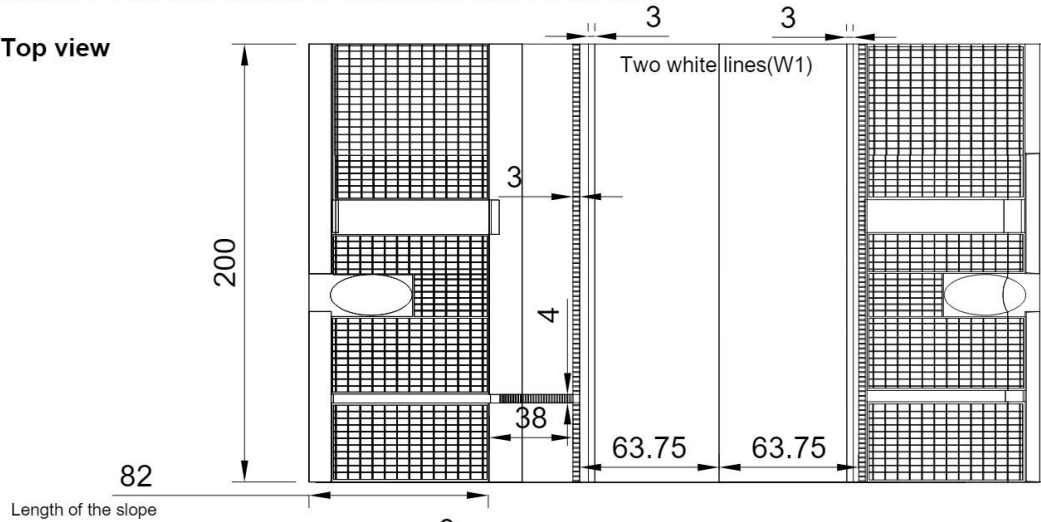
### 6.1) Model Drawing

Under this section the model is presented in the reduced format, and this will not replicate the actual size of the intended solution. The model drawing has the dimensions of 454mm, 200 mm, 155mm in X, Y and Z directions respectively, fitting within the dimensions of 500mm x 500mm x 500mm as provided in model criteria. The dimension constraint relaxation approval regarding the 200mm in the Z direction for was obtained and is attached to Appendix D.

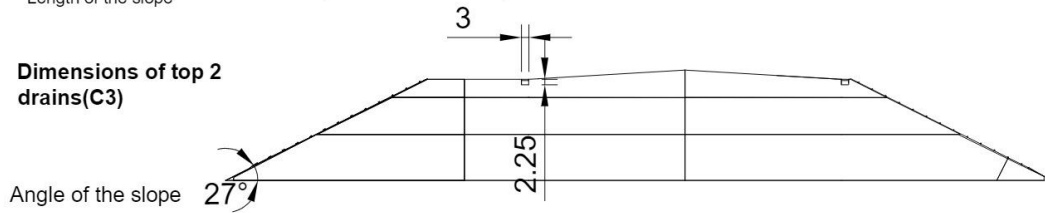


# Main Road and Pedestrian Road

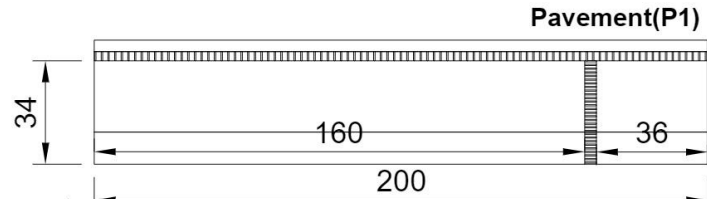
Top view



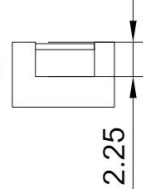
Dimensions of top 2 drains(C3)



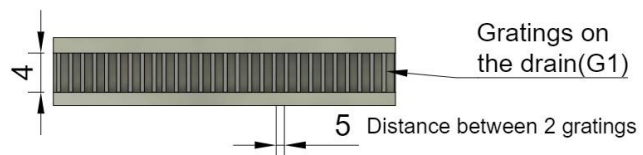
Top view of pedestrian road



Depth of the drain from the road to drain on the embankment

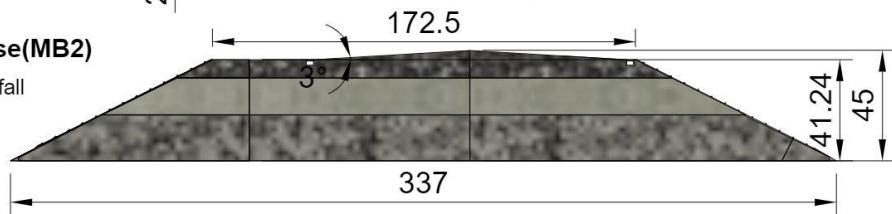


Left view



Basic Main Base(MB2)

Angle of the cross fall 3 degrees

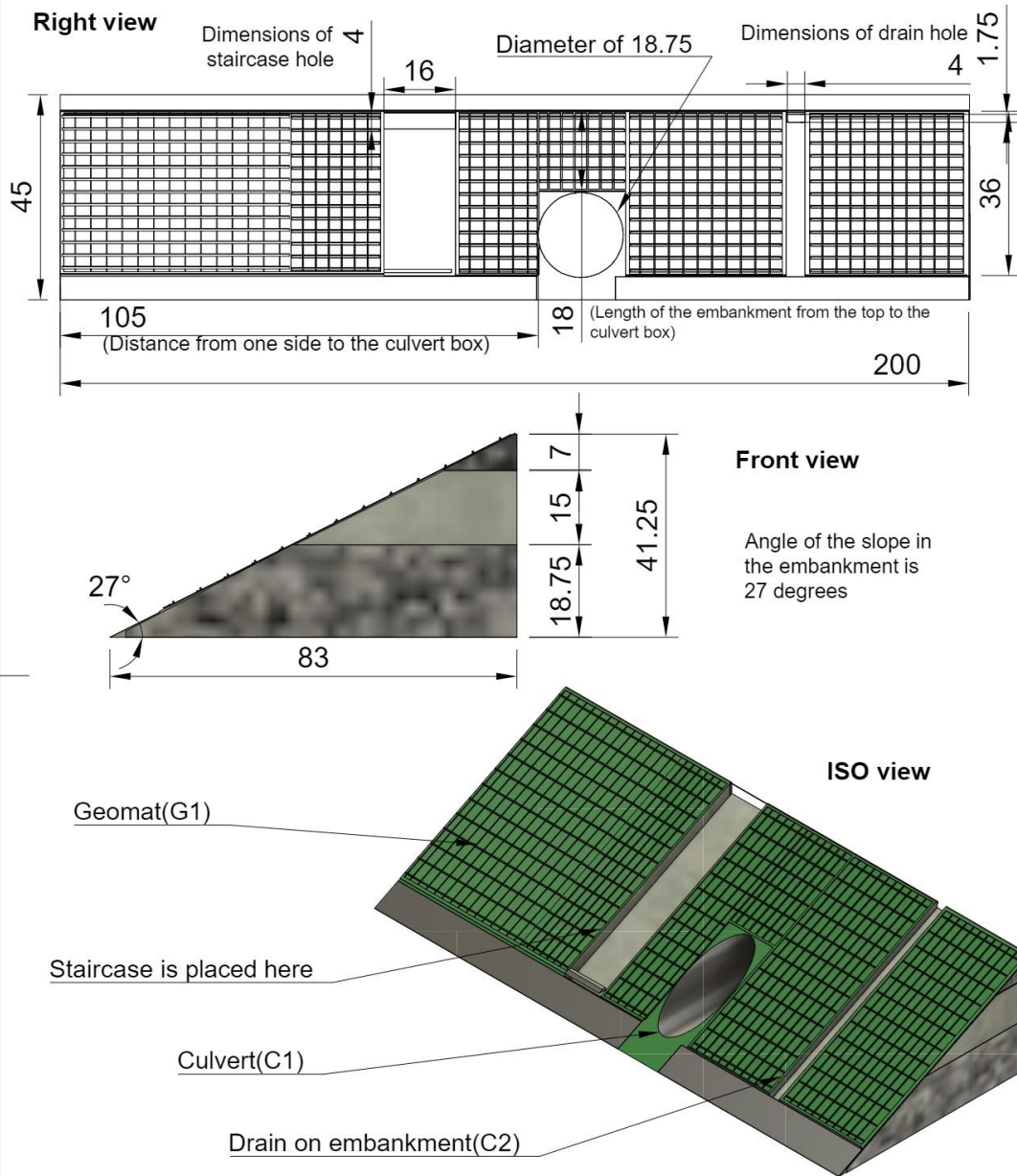


|                                                          |                     |                               |                 |               |             |
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| Dept.                                                    | Technical reference | Created by<br>Yellow Squad    | Approved by     |               |             |
| Scale<br>1 mm : 1 mm<br><br>All the dimensions are in mm |                     | Document type                 | Document status |               |             |
|                                                          |                     | Title<br><br>INDE Road Design | DWG No.         |               |             |
|                                                          |                     |                               | Rev.            | Date of issue | Sheet<br>01 |

Figure 9: Model Drawing Sheet 01 of Main Road and Pedestrian Road

# Embankment

Geomat is represented using basic square shapes drawn on the pasted A4 paper



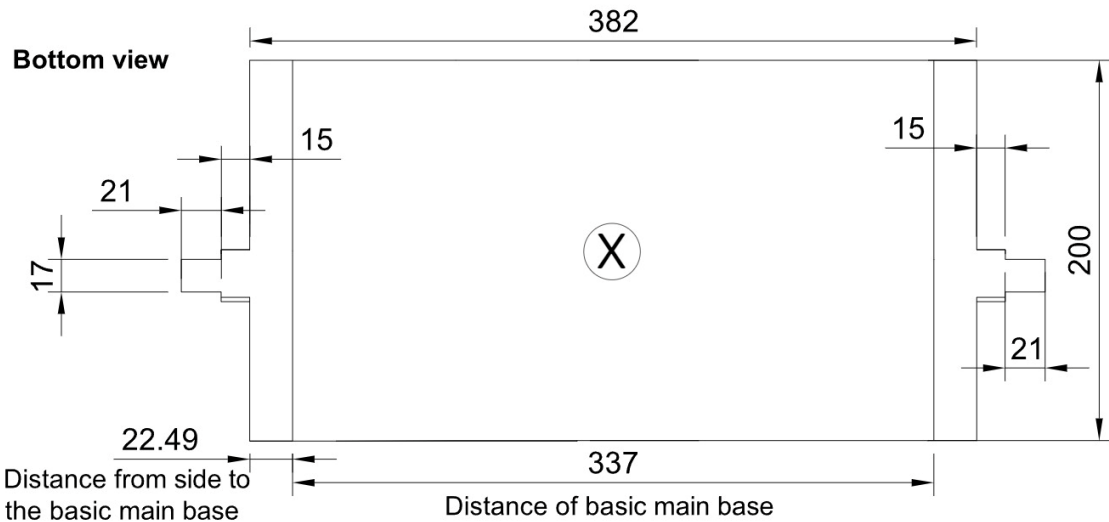
|                                                          |                     |                               |                 |               |                 |
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| Dept.                                                    | Technical reference | Created by<br>Yellow Squad    | Approved by     |               |                 |
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|                                                          |                     | Title<br><br>INDE Road Design | DWG No.         |               |                 |
|                                                          |                     |                               | Rev.            | Date of issue | Sheet<br><br>02 |

Figure 10: Model Drawing Sheet 02 of Embankment

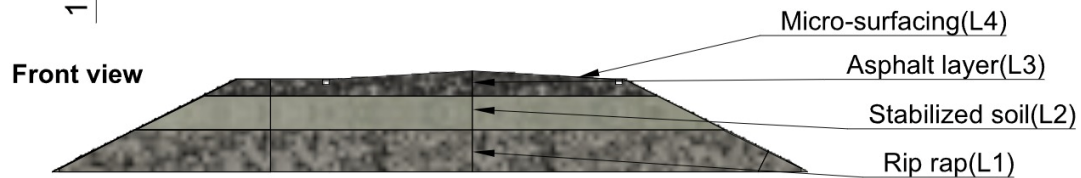
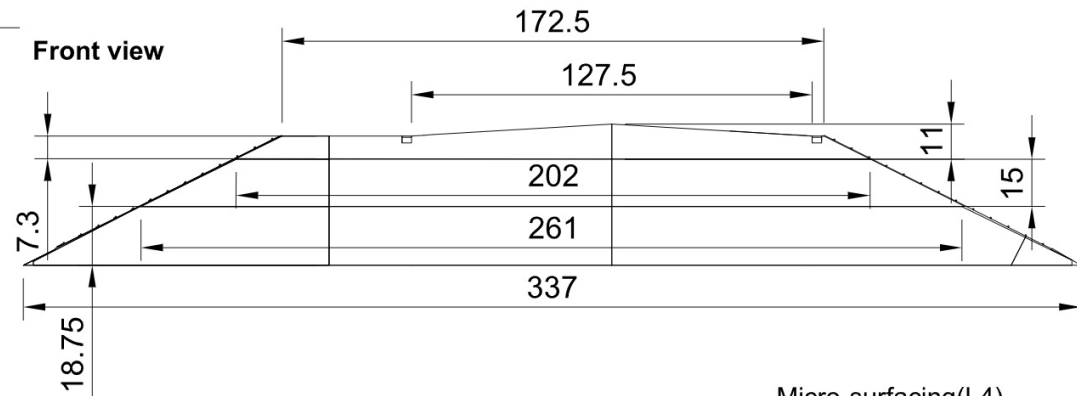
## Shape of the base and layers of the road

Shape of the base to be cut out by cardboard

Main Base Cardboard layout (Part No.01)



### Layers of road

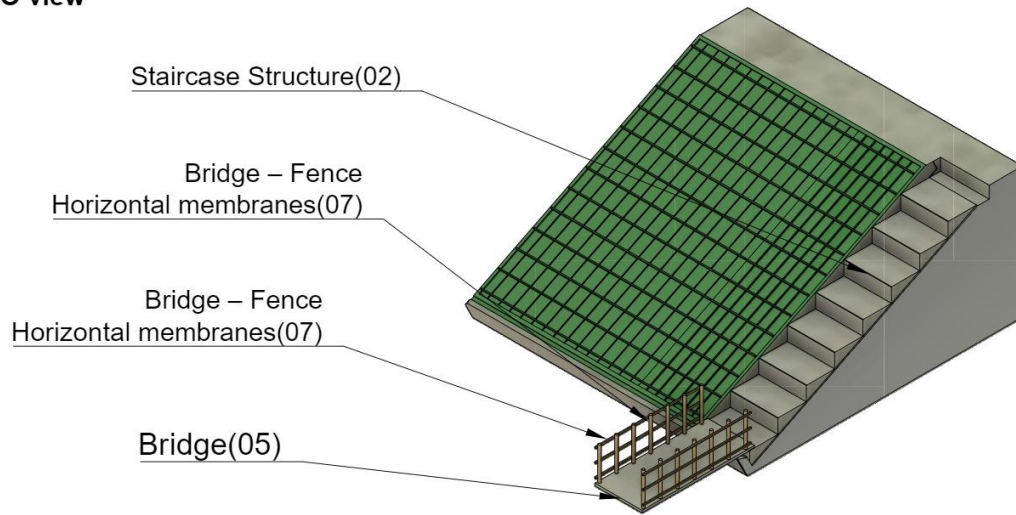


|                                                                    |                     |                                   |                 |
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| Dept.                                                              | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by     |
| <b>Scale</b><br><b>1 mm : 1 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status |
|                                                                    |                     | Title<br><b>INDE Road Design</b>  | DWG No.         |
|                                                                    |                     | Rev.                              | Date of issue   |
|                                                                    |                     | Sheet<br><b>03</b>                |                 |

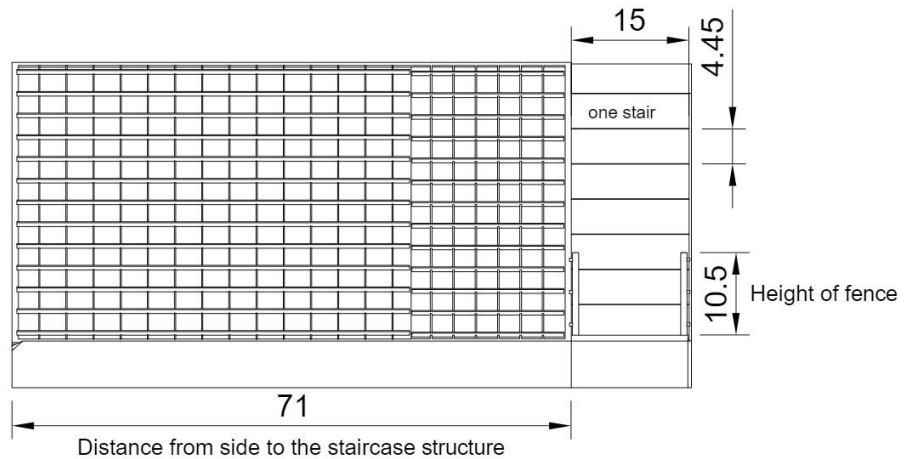
Figure 11: Model Drawing Sheet 03 of shape of base and layers of the road.

# Staircase, Bridge and Fence

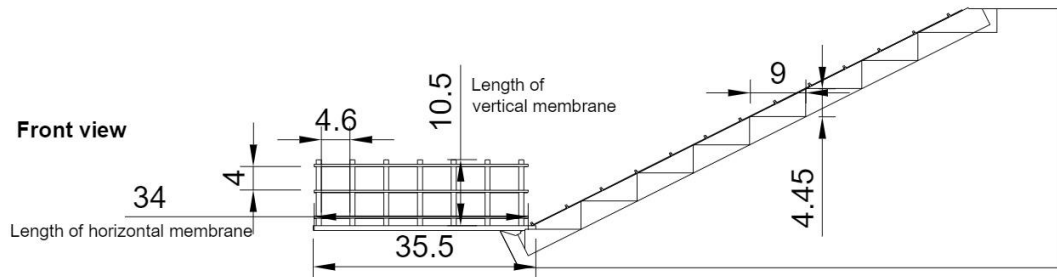
ISO view



Right view



Front view

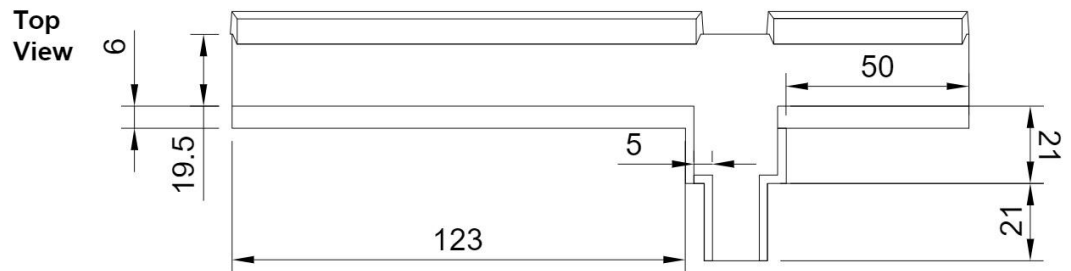


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| Dept.                                                              | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by        |
| <b>Scale</b><br><b>1 mm : 1 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status    |
|                                                                    |                     | Title<br><b>INDE Road Design</b>  | DWG No.            |
|                                                                    |                     | Rev.                              | Date of issue      |
|                                                                    |                     |                                   | Sheet<br><b>04</b> |

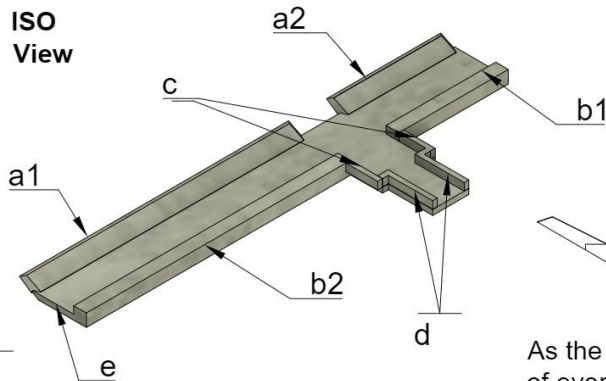
Figure 12: Model Drawing Sheet 04 of Staircase, Bridge and Fence.

# Culvert Drain

Thickness of each styrofoam that cut is 5 mm.  
Lengths and Widths are mentioned.

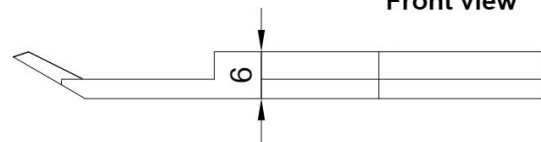


ISO View

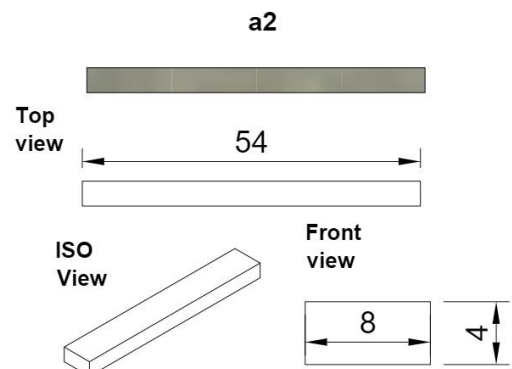
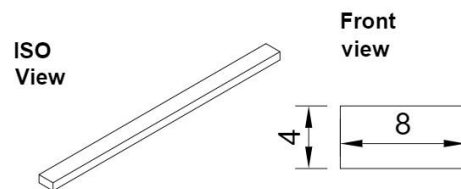
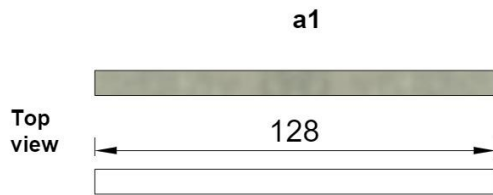


Note that e is the shape of the top view

Front view



As the A4 that is pasted is a thin layer height of every styrofoam should be 6 mm in height (except a1 & a2)



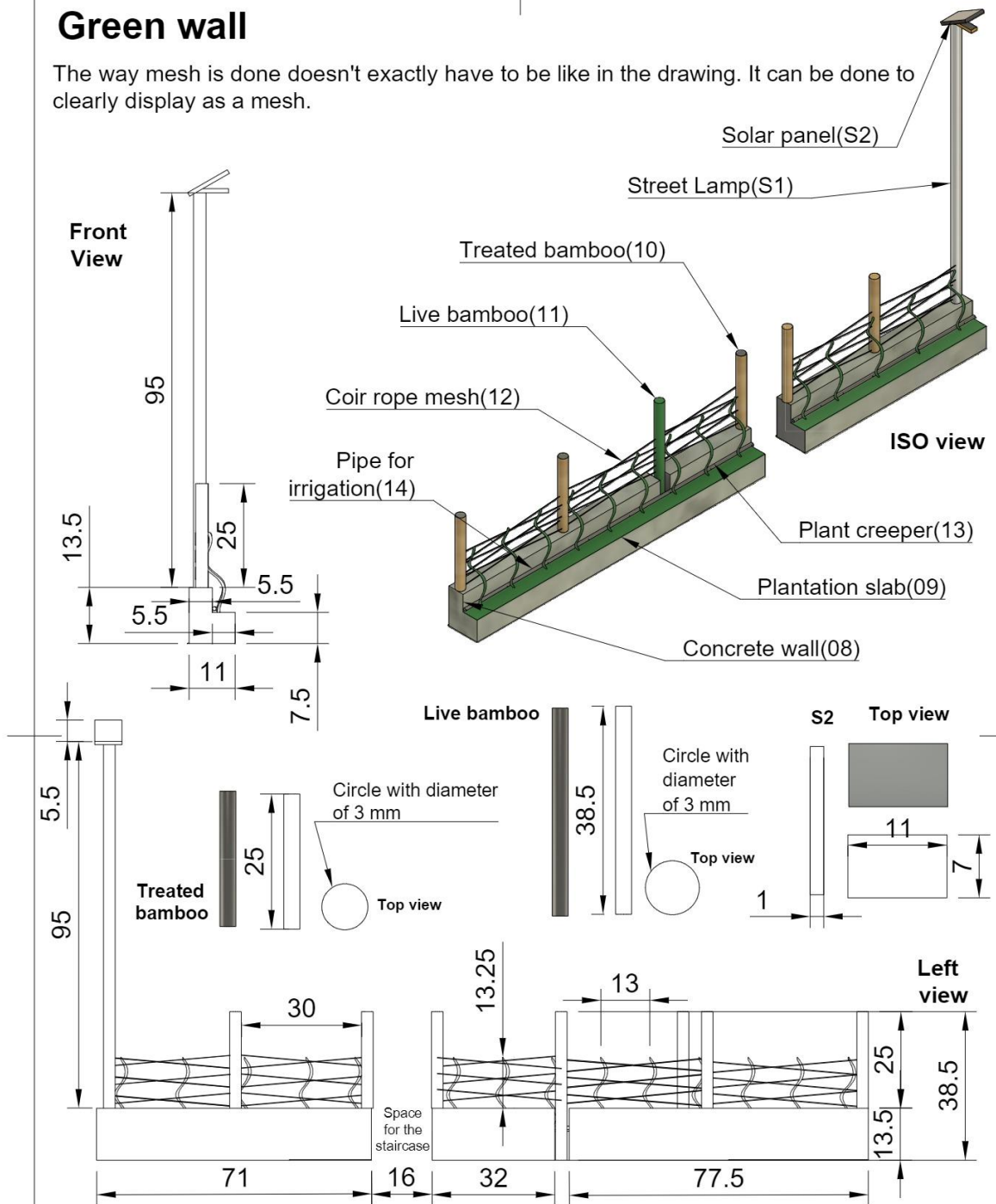
|                                                                    |                     |                                   |                 |
|--------------------------------------------------------------------|---------------------|-----------------------------------|-----------------|
| Dept.                                                              | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by     |
| <b>Scale</b><br><b>1 mm : 1 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status |
|                                                                    |                     | Title<br><b>INDE Road Design</b>  | DWG No.         |
|                                                                    |                     | Rev.                              | Date of issue   |
|                                                                    |                     | Sheet                             | <b>05</b>       |

Figure 13: Model Drawing Sheet 05 of Culvert Drain.



## Green wall

The way mesh is done doesn't exactly have to be like in the drawing. It can be done to clearly display as a mesh.



|                                                                    |                     |                                   |                 |
|--------------------------------------------------------------------|---------------------|-----------------------------------|-----------------|
| Dept.                                                              | Technical reference | Created by<br><b>Yellow Squad</b> | Approved by     |
| <b>Scale</b><br><b>1 mm : 1 mm</b><br>All the dimensions are in mm |                     | Document type                     | Document status |
|                                                                    |                     | Title<br><b>INDE Road Design</b>  | DWG No.         |
|                                                                    |                     | Rev.                              | Date of issue   |
|                                                                    |                     | Sheet<br><b>06</b>                |                 |

Figure 14: Model Drawing Sheet 06 of Green wall.

## 6.2) Specifications

In this section, the process involved in the creation of a scaled-down representative model of the proposed design will be outlined.

### 6.2.1) Bill of Quantities (and Supplier)

Table 10: Bill of materials for the model construction

| Description                                                      | Material                 | Source                                                     | Quantity                                                                                              | Cost per item | Sub Total |
|------------------------------------------------------------------|--------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------|-----------|
| Cuboid Styrofoam (Diameter x Thickness) 19" by 6"                | Styrofoam Cake Structure | Hants No. 191, Hekitta Rd, Wattala, Sri Lanka              | 1                                                                                                     | AUD 10.62     | AUD 10.62 |
| Cuboid Styrofoam (Diameter x Thickness) 12" by 4"                | Styrofoam Cake Structure | Hants No. 191, Hekitta Rd, Wattala, Sri Lanka              | 1                                                                                                     | AUD 2.99      | AUD 2.99  |
| Corrugated Carton 1 kg, with an unfolded length of at least 50cm | Cardboard                | Paper Story Lanka<br>79 Galwala Rd, Nugegoda 10250         | 1kg                                                                                                   | AUD 1         | AUD 1     |
| Colored A4 Colours                                               | A4 Papers                | Local bookshops and stationery shops (i.e. Wisdom)         | Grey – 2<br>Brown – 1<br>Dark Grey – 2<br>Light green – 4<br>Light grey – 4<br>Black - 2<br>White - 1 | AUD 0.10      | AUD 1.6   |
| CHEMIFIX GLUE (100g)                                             | Glue                     | Local bookshops (i.e. Wisdom)                              | 1                                                                                                     | AUD 1.76      | AUD 1.76  |
| B.B.Q Sticks Pack 10 inches                                      | B.B.Q Sticks             | BakersCraft<br>44, St John's Road<br>Colombo 11, Sri Lanka | 1                                                                                                     | AUD 1.27      | AUD 1.27  |
| 0.5 mm thickness                                                 | Binding wire             | Local Hardware Shop                                        | 1                                                                                                     | AUD 0.5       | AUD 0.5   |

|                                                                                          |                |                                                             |                        |             |              |
|------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------|------------------------|-------------|--------------|
| 2 mm thickness                                                                           | Twine thread   | Local bookshops and stationary shops<br>(i.e. Wisdom)       | Green – 1<br>White - 1 | AUD<br>0.5  | AUD<br>1     |
| White colour drinking straw. Thickness must be 5 mm and length should be at least 101 mm | Drinking Straw | Local grocery shops and stationary shops                    | 1                      | AUD<br>0.1  | AUD<br>0.1   |
| ~350gsm                                                                                  | Box board      | Local bookshops and stationary shops<br>(i.e. M.D.Gunasena) | 1                      | AUD<br>0.73 | AUD<br>0.73  |
| Total Cost                                                                               |                |                                                             |                        |             | AUD<br>21.57 |

### 6.2.2) Tools List

1. A box cutter
2. Paper cutter pen
3. An iron stick (Diameter around 2 cm)
4. 2 Platignum pens (Green and Black)
5. Pencil
6. Ruler
7. Protractor
8. Scissors
9. Compass
10. Sandpaper
11. Wire Cutter



## 6.2.3) Methodology Steps

### 6.2.3.1) Parts Specification

#### 6.2.3.1.1) Shapes to be cut

##### Main Base Cardboard layout (Part No.01)

1. Mark the cutting outline using the Platignum marker pen on the cardboard referring to the outermost border of the bottom view of the base of the whole model (Sheet 3, Part No.01).
2. Cut the cardboard according to the outline in step 1 using the box cutter.

##### Basic Main Base (Part No.MB2)

1. Mark the cutting lines on the sides and top of the main Styrofoam cuboid using the Platignum marker pen by referring to (Sheet 1, Part No.MB2) and the bottom view of the main base (Sheet 3, Part No.01). Use a protractor to measure the angle specified area in (Sheet 2, Front view)
2. Using a box cutter, cut the Styrofoam cuboid according to the cutting lines marked in step 1.
3. Polish the surfaces if necessary, using sandpaper such that the cut surfaces are smooth.

##### Drainage Cuts on the Embankment (Part No.C2)

1. Mark the cutting lines on the sides and top of the Styrofoam cuboid using the Platignum marker pen by referring to (Sheet 2, Right view) and (Sheet 1, Top view) such that two culvert drain cuts are made on both sides of the embankment as shown in the (Sheet 2, Front view). The depth of the cut should be 3 mm.
2. Using the paper cutter pen (or box cutter if needed), cut it accordingly. (For easiness, cut it part by part.)
3. Use sandpaper to smooth it if the edges are rough.

##### Top Two Drain Cuts (Part No. C3)

1. Mark the cutting lines on the sides and top of the Styrofoam cuboid using the Platignum marker pen by referring to (Sheet 1, Top view) such that two drain cuts are made on both sides of the road as shown in (Sheet 1, Dimensions of top 2 drains(C3)).

2. Using the paper cutter pen (or box cutter if needed), cut it accordingly. (For easiness, cut it part by part.)
3. Use sandpaper to smooth it if the edges are rough.
4. Colour the inside of each drain cuts with the black using the black colour Platignum.

#### Two Staircase cuts on the Embankment (Part No.02)

1. Mark the cutting lines on the sides and top of the Styrofoam cuboid using the Platignum marker pen by referring to the (Sheet 2, Right view) such that two cuts are made on both side of the embankment as shown in the (Sheet 1, Top view). The depth of the cut should be 4 mm
2. Using the paper cutter pen (or box cutter if needed), cut it accordingly. (For easiness, cut it part by part.)
3. Use sandpaper to smooth it if the edges are rough.

#### Culvert Drainage Shapes

1. Mark the cutting lines on the sides and top of the other Styrofoam cuboid for each shape by referring to the dimensions of each part of the side and top views of each mentioned part in table 11.

*Table 11: Culvert drainage Styrofoam shapes*

| <b>Part No</b> | <b>Location of Dimensions in Technical Drawing</b> | <b>Quantity Needed</b> |
|----------------|----------------------------------------------------|------------------------|
| <b>a1</b>      | (Sheet 5, a1, Top view & Front view)               | 2                      |
| <b>a2</b>      | (Sheet 5, a2, Top view & Front view)               | 2                      |
| <b>b1</b>      | (Sheet 5, Top view & Front view)                   | 2                      |
| <b>b2</b>      | (Sheet 5, Top view & Front view)                   | 2                      |
| <b>c</b>       | (Sheet 5, Top view & Front view)                   | 4                      |
| <b>d</b>       | (Sheet 5, Top view & Front view)                   | 4                      |

2. Using the paper cutter pen (or box cutter if needed), cut it accordingly

3. Use sandpaper to smooth it if the edges are rough.

#### Bridge (Part No.05)

1. Mark the cutting lines on the sides and top of a Styrofoam cuboid by referring to the dimensions of the (Sheet 4, Right view & Front view). The thickness of the bridge is 5 mm.

2. Using the paper cutter pen (or box cutter if needed), cut it accordingly.

3. Use sandpaper to smooth it if the edges are rough.

4. Repeat steps 1 to 3 to make another bridge for the other side.

#### Concrete wall (Part No.08) and plantation slab (Part No.09) for Green Wall

1. Mark the cutting lines on the sides and top of a Styrofoam cuboid for each shape by referring to the dimensions of each part of the views of each mentioned part in table 12.

*Table 12: Green wall Styrofoam shapes*

| Part                                | Location of Dimensions in Technical Drawing | Quantity Needed                                                                                                  |
|-------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Concrete wall (Part No.08)</b>   | (Sheet 6, Left view & Front view)           | <b>2:</b> Note that the concrete wall is divided into parts, so each of their dimensions should be considered.   |
| <b>Plantation Slab (Part No.09)</b> | (Sheet 6, Left view & Front view)           | <b>2:</b> Note that the plantation slab is divided into parts, so each of their dimensions should be considered. |

2. Using the paper cutter pen (or box cutter if needed), cut it accordingly.

3. Use sandpaper to smooth it if the edges are rough.

### BBQ Stick shapes

1. Mark the cutting lines on the side of each BBQ stick required by referring to the dimensions of each part in table 13.

*Table 13: BBQ stick parts*

| <b>Part</b>                                             | <b>Length to be cut</b>          | <b>Quantity Needed</b> | <b>Special Instructions</b>                                                                                            |
|---------------------------------------------------------|----------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Green Wall-Treated Bamboo (Part No.10)</b>           | Sheet 6, Treated Bamboo          | 5                      | Take 10mm more than the mentioned height in sheet 6 to insert into Styrofoam.                                          |
| <b>Green Wall-Live Bamboo (Part No.11)</b>              | Sheet 6, Live Bamboo             | 1                      | Take 10mm more than the mentioned height in sheet 6 to insert into Styrofoam.<br>Color with green platignum marker pen |
| <b>Bridge - Fence Vertical membranes (Part No.06)</b>   | Sheet 4, Right view & Front view | 28                     | None                                                                                                                   |
| <b>Bridge – Fence Horizontal membranes (Part No.07)</b> | Sheet 4, Right view & Front view | 12                     | Color with black platignum marker pen                                                                                  |

2. Using the paper cutter pen (or box cutter if needed), cut it accordingly.

### Gratings (Part No. G2)

1. Mark the cutting lines on binding wire required by referring to the dimensions mentioned in (Sheet 1, Left view & Top view) such that 88 gutter pieces are made.

2. Using the Wire cutter, accordingly, cut them.

### Culvert (Part No.C1)

1. Mark the two circles on either side of the Styrofoam cuboid using a pencil and a compass with the diameter taken as mentioned in (Sheet 2, Right view). Ensure the cutting lines are made correctly so that the culvert's location is correct.

2. Heat the iron stick by exposing it to the flame of a gas stove.

3. Insert and take along the circle drawn in step 1 until the whole culvert part is perforated accordingly.

4. If the edges are too rough and not neat, use sandpaper to smoothen the surface.

### Twine thread Cuts

1. Using the ruler, cut twine thread pieces using the scissors as required in table 14.

*Table 14: Twine thread parts*

| <b>Part</b>                            | <b>Length</b>                                                                        | <b>Color</b> | <b>Quantity</b>                                           |
|----------------------------------------|--------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------|
| Green wall – Coir Mesh<br>(Part No.12) | 30mm : Note<br>additional 5mm can<br>be taken to clear<br>show the mesh.             | White        | As required to clearly<br>show the mesh like<br>structure |
| Green wall – Plant<br>Creeper          | 27 mm : Note<br>additional 5mm can<br>be taken to clear<br>show thread is<br>curved. | Green        | 14                                                        |

### Staircase (Part No.02) structure

1. Draw the cutting lines and folding lines on the box board with reference to the dimensions mentioned in Sheet 4, Right view & Front view for Staircase. Note that the cutting lines are the main rectangular shape, and the folding lines can be drawn in the middle such that dimensions mentioned in Sheet 4 are achieved when folded.

2. Repeat step 1 so that 2 structures are made.

3. Fold all 2 cut pieces following the folding lines, such that the staircase structure is made.

#### Irrigation Pipe in Green Wall (Part No. 14)

1. By referring to (Sheet 6, Part No. 14), draw a 2 mm thick straight line using the black colour platignum marker pen so that the pipe is clearly visible.

#### Street Lamp (Part No. S1)

1. Draw cutting lines on the straw using a black platignum marker pen to match the dimensions in (Sheet 06, Front View, S1).

2. Cut it accordingly using scissors.

3. Fold the top joint part of the straw as such to achieve the horizontal length mentioned in (Sheet 06, Front View, S1).

#### **6.2.3.1.2) Pieces of A4s, for specific parts**

1. Measure and draw the outlines using a pencil for each of the required parts mentioned in the table 15 by referring to the dimensions in the corresponding technical drawing and cut each piece in the mentioned quantity using scissors (Box cutter or paper cutter pen if needed).

*Table 15: Pieces of A4s to be cut.*

| <b>Part</b>                               | <b>Location of Dimensions<br/>in Technical Drawing</b> | <b>Colour<br/>of A4</b> | <b>Quantity</b> | <b>Special Instructions</b>                               |
|-------------------------------------------|--------------------------------------------------------|-------------------------|-----------------|-----------------------------------------------------------|
| Cross Section<br>Layer 1 (Part No.<br>L1) | Sheet 3, Layers of road –<br>Front view                | Grey                    | 2               | Use the protractor to measure<br>the angle of 27 degrees. |
| Cross Section<br>Layer 2 (Part No.<br>L2) | Sheet 3, Layers of road –<br>Front view                | Brown                   | 2               |                                                           |
| Cross Section<br>Layer 3 (Part No.<br>L3) | Sheet 3, Layers of road –<br>Front view                | Dark<br>Grey            | 2               |                                                           |

|                                    |                                                          |             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|----------------------------------------------------------|-------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geomat on embankment (Part No. G1) | Sheet 2, Right view                                      | Light Green | 2 | <p>- To get the needed shape, mark the cutting lines and remove areas of the culvert circle, drainage and staircase referring to each dimension from Sheet 2, right view.</p> <p>- Draw lines using the green Platignum on the cut light green A4 such that square shapes are made as shown in sheet 02, G1. Note that the model drawing squares are for representation only, and it is ideal to have squares with a length of 5mm.</p> |
| Pavement (Part No. P1)             | Sheet 1, Top view of Pedestrian Road                     | Light Grey  | 1 | None                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Road Surface (Part No. L4)         | Sheet 3, Layers of road – Front view & Sheet 1, Top View | Black       | 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Bridge (Part No. 05)               | Sheet 4, Right view & Front view                         | Light Grey  | 2 | Cut accordingly such that top and side areas are covered.                                                                                                                                                                                                                                                                                                                                                                               |
| Culvert Drain (Part a1)            | Sheet 5, a1, Top view & Front view                       |             | 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Culvert Drain (Part a2)            | Sheet 5, a2, Top view & Front view                       |             | 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Culvert Drain (Part b1)            |                                                          |             | 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Culvert Drain (Part b2)            |                                                          |             | 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                                  |                                            |               |   |                                                                                         |
|------------------------------------------------------------------|--------------------------------------------|---------------|---|-----------------------------------------------------------------------------------------|
| Culvert Drain<br>(Part c)                                        | Sheet 5, Top view &<br>Front view          | Light<br>Grey | 4 |                                                                                         |
| Culvert Drain<br>(Part d)                                        |                                            |               | 4 |                                                                                         |
| Culvert Drain<br>(Part e)                                        |                                            |               | 2 |                                                                                         |
| Green Wall -<br>Concrete wall<br>(Part No. 08)                   | Sheet 6, Left view &<br>Front view         |               | 2 | Cut accordingly such that top<br>and side areas are covered.                            |
| Green Wall -<br>Plantation slab<br>(Part No. 09) -<br>side layer | Sheet 6, Left view &<br>Front view         |               | 2 |                                                                                         |
| Green Wall -<br>Plantation slab<br>(Part No. 09) -<br>top layer  | Sheet 6, Left view &<br>Front view         |               | 2 |                                                                                         |
| Drain on<br>embankment<br>(Part No.C2)                           | Sheet 2, Right view &<br>Sheet 1, Top view | Light<br>Grey | 2 | When cutting, make sure all<br>area inside the drain on<br>embankments cut are covered. |
| White Lines on<br>the Road (W1)                                  | Sheet 1, Top view                          | White         | 2 | None                                                                                    |
| Solar Panel(Part<br>No.S2)                                       | Sheet 6, Top view                          | White         | 1 | Should be pasted on top of the<br>Street lamp straw(S1) as shown<br>in Sheet 6.         |

2. After cutting each piece specified, paste each cut A4 shape on their respective locations as mentioned in the first 2 columns of the above table using Chemfix glue. (e.g. Cross Section Layer 1 (Part No. L1) should be pasted on the basic main base Styrofoam in its respective locations as shown in (Sheet 03)).



Note: Still the A4 pieces are pasted on separate Styrofoam shapes. (i.e. Parts associated with the main base Styrofoam shape, Culvert Drainage Styrofoam shapes, Bridge Styrofoam shape, Green wall concrete wall and plantation slab Styrofoam shapes).

### 6.2.3.2) Assembly Specification

1. Fix each of the mentioned parts in their respective locations with a pencil as mentioned in the table 16. For convenience, measure and mark the locations where each of the parts should be fixed (i.e. the border of places where each component, like the main base and culvert drainage parts, needs to be fixed on the Main Base cardboard layout, which can be marked according to the technical drawing before fixing).

Note that Assembly specifications are given by referring to both concept drawings and model drawings.

Table 16: Specifications on assembling parts of the model

| Part                             | Location to be fixed                                                                                    | Method of fixing   | Special Instructions                                                |
|----------------------------------|---------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------|
| Basic Main Base (Part No.MB2)    | To Main Base Cardboard layout (Part No. 01) – according to Sheet 03, rectangle mentioned with Symbol X. | Glue using Chemfix | Mark the gluing places using a pencil before fixing.                |
| Culvert Drain (Part a1)          | To Main Base                                                                                            |                    | Same procedure to the other side as well.                           |
| Culvert Drain (Part a2)          | Cardboard layout (Part                                                                                  |                    |                                                                     |
| Culvert Drain (Part b1)          | No. 01)- according to                                                                                   |                    |                                                                     |
| Culvert Drain (Part b2)          | Sheet 05, Top view                                                                                      |                    |                                                                     |
| Culvert Drain (Part c)           |                                                                                                         |                    |                                                                     |
| Culvert Drain (Part d)           |                                                                                                         |                    |                                                                     |
| Culvert Drain (Part e)           |                                                                                                         |                    |                                                                     |
| Staircase structure (Part No.02) | To Basic Main Base (MB2) – according to                                                                 |                    | Add lines of Chemfix glue on the sides of the staircase cut already |

|                                                  |                                                                                           |                                                                                                                                                                                |                                                                                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | Sheet 04, Right and Front view                                                            |                                                                                                                                                                                | made in the main base Styrofoam.<br><br>Afterwards fix the staircase structure such that the folded nature is not gone.                   |
| Gratings (Part No.04)                            | To Basic Main Base (MB2) – according to Sheet 01, Top view of pedestrian road             | Insert the pre-cut aluminum wire pieces to the top drain cuts made in the main base Styrofoam by inserting one end of the wire to one side of the Styrofoam and then the other | Before inserting the pieces, mark the locations of the gutters to be fixed using a pencil by referring to the gap length in the Sheet 01. |
| Bridge (Part No.05)                              | To Culvert Drain (Part a2) and Culvert Drain (Part b1) – according to Sheet 2, Front View | Glue using Chemfix                                                                                                                                                             | None                                                                                                                                      |
| Bridge - Fence Vertical membranes (Part No.06)   | To Bridge (Part No. 05) – according to Sheet 04, Front View                               | Insert to the Styrofoam by pressing                                                                                                                                            | Measure and mark the location of each fence place using a pencil before inserting.                                                        |
| Bridge – Fence Horizontal membranes (Part No.07) | To Bridge – Fence Vertical membranes (Part No. 06) – according to Sheet 04, Front view    | Glue it by using Chemfix so that they are fixed horizontally as shown in Sheet 04.                                                                                             | Measure and mark the location of each fence place using a pencil before gluing.                                                           |
| Green Wall - Concrete wall (Part No.08)          | To Basic Main Base (MB2) – according to                                                   |                                                                                                                                                                                | None                                                                                                                                      |

|                                               |                                                                                                                                                                                          |                                      |                                                                                                                                                     |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | Sheet 01, 28.5 mm from the gratings                                                                                                                                                      |                                      |                                                                                                                                                     |
| Green wall – Plantation slab (Part No.09)     | <p>To Basic Main Base (MB2) – according to Sheet 01, 23 mm from the gratings</p> <p><b>AND</b></p> <p>To Green Wall - Concrete wall (Part No.08) – according to Sheet 06, Front view</p> | Glue using Chemfix                   | Note that the plantation slab shape should be glued such that it should be glued to both the top part of basic main base and side of concrete wall. |
| Green wall – Treated bamboo (Part No.10)      | To Green Wall - Concrete wall (Part No.08) – according to Sheet 06, Left view                                                                                                            | Insert to the Styrofoam by pressing. | Measure and mark the location of each place using a pencil before inserting.                                                                        |
| Green wall – Live bamboo (Part No.11)         | To Basic Main Base (MB2) – according to Sheet 06                                                                                                                                         |                                      | None                                                                                                                                                |
| Solar Street Lamp(S1) with (S2) fixed on top. | To Green Wall - Concrete wall (Part No.08) – according to Sheet 06, Left view                                                                                                            |                                      | None                                                                                                                                                |

|                                            |                                                                                                                                                                         |                    |                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Green wall – Coir Mesh<br>(Part No.12)     | To Green Wall –<br>Treated bamboo (Part<br>No.10) , Live Bamboo<br>(Part No.11) and Street<br>Lamp(Part No.S1) –<br>according to Sheet 06,<br>Left View                 | Glue using Chemfix | - Fix each twine thread<br>accordingly to achieve<br>a mesh- like structure.                                                                                                                                                                                                                                                                           |
| Green wall – Plant<br>Creeper (Part No.13) | To Green Wall – Coir<br>Mesh (Part No.12)-<br>According to Sheet 06<br><br><b>AND</b><br><br>Green Wall –<br>Plantation slab (Part<br>No.09) - According to<br>Sheet 06 |                    | - The top part of the<br>twine thread should be<br>pasted on the twine<br>thread of the coir<br>mesh whereas the<br>other end of the twine<br>thread should be<br>pasted to the top<br>brown side of the<br>Green Wall –<br>Plantation slab(09).<br><br>- Fix each twine thread<br>according to the length<br>gap mentioned in<br>Sheet 06, Left view. |

#### 6.2.4) Testing and Quality

**Enduring Test:** The completed model should be able to withstand rigidly when kept on a plain surface. Fairly move the handheld model to the left and right 5 times and upward and downward 5 times to ensure that all the parts are fixed without any loose components.

**Dimensions Quality Test:** Ensure that all the model components are aligned with the dimensions mentioned in the specifications and model drawings. An error of  $\pm 3 \text{ mm}$  can be tolerated.

## 7.0) Tender Evaluation Criteria

### 7.1) Previous Experience

This model design construction includes activities like cutting and gluing Styrofoam precisely, using heated objects to perforate Styrofoam precisely, sticking small items like toothpicks, crafting using cardboard and overall being skilful in precise handcrafts. Therefore, a company with contractors who have prior experience and who have done previous projects regarding those categories has an added advantage.

**a)** Below are **categories of handcrafts** accepted in this criterion:

Successful completion of handcrafts which consist of,

- i) using tools like compasses, markers, rulers, box cutters and glues to handcraft precisely
- ii) precision cutting, gluing and perforating materials of Styrofoam (or similar) and cardboard.

(In this section(a) **successful completion** of handcraft is defined as completing the handcraft meeting the minimum required standard of expectation of such handcraft)

**b)** The items below are accepted **as evidence** for factors mentioned in the scoring table,

- i) Photograph(s)/video(s) of such mentioned handcrafts taken on a date after completion of it.  
(Each contractor is expected to show proof of the exact date when the photograph/video was taken) (**OR**)
- ii) Proof of certificate of participation/ achievement if the contractor has won or participated in a competition/event that involves the handcrafts mentioned.

Scores will be allocated according to the table 17.

Table 17: Previous Experience score

| Criteria                                                                                            | Score |
|-----------------------------------------------------------------------------------------------------|-------|
| All the team members have done handcrafts which consists of both categories a) (i) and (ii)         | 5     |
| Three team members have done handcrafts, which consists of both categories a) (i) and (ii)          | 4     |
| One team member has done handcrafts, which consists of both categories a) (i) and (ii)              | 2     |
| One team member has done handcrafts, which consists of the category a) (i)                          | 1     |
| None of the team members has done handcrafts, which consists of either categories of a) (i) or (ii) | 0     |

## 7.2) Teamwork Experience

Considering the short timeframe to complete this model construction, effective teamwork is vital to finish it on time. Therefore, a company with contractors who have previous teamwork experience will gain an advantage.

**a)** Below are **categories of teamwork** accepted in this criterion:

Successful participation in a group activity/activities of

- i) of team-based sports.
- ii) team-based learning activities (ex: Educational group assignments)
- iii) team-based competitions including activities like handcrafts.

(Here **successful participation** is defined as completing the group activity/ activities with no penalties or warnings by supervisors due to team rule violation)

**b)** The items mentioned below are accepted **as evidence** for factors mentioned in the scoring table,

- i) Proof of certificate of participation/achievement in mentioned activities. **(OR)**
- ii) Photograph(s)/video(s) which shows participation in mentioned activities **(OR)**

- iii) Photograph(s)/screenshot(s) of successful submission of group work of mentioned activities.

Scores will be allocated according to the table 18.

Table 18: Teamwork Experience score

| Criteria                                                                                                                                     | Score |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------|
| All team members have successfully participated in at least one activity of <b>one</b> category of a) (i) <b>or</b> (ii) <b>or</b> (iii)     | 5     |
| Four team members have successfully participated in at least one activity of <b>one</b> category of a) (i) <b>or</b> (ii) <b>or</b> (iii)    | 4     |
| Three team members have successfully participated in at least one activity of <b>one</b> category of a) (i) <b>or</b> (ii) <b>or</b> (iii)   | 3     |
| Two team members have successfully participated in at least one activity of <b>one</b> category of a) (i) <b>or</b> (ii) <b>or</b> (iii)     | 2     |
| None of the team members have successfully participated in at least one activity from <b>any</b> of the categories in a) (i), (ii) and (iii) | 0     |

### 7.3) Availability of Tools

Since the construction model design must be done briefly, all necessary tools should be available and ready. If the necessary tools are not available, it will be time-consuming to go around searching for them and the team won't have enough time to construct the model

a) Tools needed are (Specified in the tools section under Specifications):

- 1) Box cutter, paper cutter pen, wire cutter and scissors- to cut materials like cardboard, binding wire, A4 and Styrofoam
- 2) Ruler, Protractor, and compass - to gain the accurate measurement required
- 3) 2 Platignum marker pens and pencil- to help mark measurements and color BBQ sticks
- 4) Iron rod (~ diameter 2 cm)- to perforate Styrofoam
- 5) Sandpaper – To smoothen the rough surfaces

**Evidence-** photographs or videos of the needed tools

Scores will be allocated according to the table 19.

*Table 19: Availability of Tools score*

| Criteria                                                                           | Score |
|------------------------------------------------------------------------------------|-------|
| Have all the needed tools                                                          | 5     |
| Have eight tools including a) all of 1) <b>and</b> a)4 and a)5 of the needed tools | 3     |
| Have five tools including a)4) and a)5 of the needed tools                         | 2     |
| Do not have any of the needed tools                                                | 0     |

## 7.4) Availability for Meetings

Since the time frame is small, all the team members should have the ability to attend all the meetings without any absence. A company with a proven track record of members with a low absent rate has an added advantage.

**Evidence** – Previous Weekly Progress Reports of stage 1.

Scores will be allocated according to the table 20.

*Table 20: Availability for Meetings score*

| Criteria                                                             | Score |
|----------------------------------------------------------------------|-------|
| None of the team members have a track record of being absent in WPR  | 5     |
| Two of the team members have a track record of being absent in WPR   | 3     |
| Three of the team members have a track record of being absent in WPR | 2     |
| All the team members have a track record of being absent in WPR      | 0     |



## 8.0) Conclusion

The design package proposes a comprehensive solution to improving road access during the wet season on Saibai Island. The challenge has been addressed according to the stakeholder needs and selected design criteria and constraints.

Our proposed concept focuses on creating safe, durable, easy-to-maintain roads, even in adverse weather conditions. The issue of flooding the road with rain and tidal waves is addressed with a road pavement design and a drainage system. The road was elevated by 1.1m while utilizing a layered approach with materials such as riprap, stabilized soil, reclaimed asphalt pavement, and micro-surfacing to ensure the durability and longevity of the road. For the structural stability of the elevated road, embanks are implemented with the soil erosion control measure, geomats. Incorporating road cross-fall and roadside drains to facilitate proper water runoff and prevent pavement defects ensures the longevity of the road infrastructure.

With the road pavement design and drainage system enhancing flood control, the pedestrian road is designed with concrete pavement and safety features such as green walls to enhance pedestrian safety and durability. Apart from pedestrian safety, the design of the green wall introduces a unique combination of live and treated bamboo along with hydrogel crystals as a further erosion control and to improve air quality and optimize water usage, thereby offering additional benefits to the community.

In conclusion, our design package offers a holistic approach to addressing the challenges faced by Saibai Island, providing a sustainable and resilient solution that prioritizes the well-being and safety of its residents. By carefully considering various factors and innovative design elements, we aim to positively impact the island's infrastructure and contribute to its overall development and prosperity.

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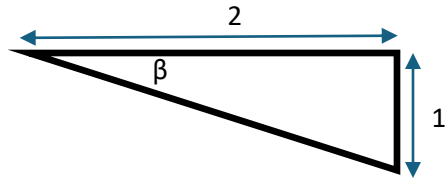
## 10.0) Appendices

### Appendix A – Calculations

#### Angle of the Embankment

Slope of the embankment= 1:2(V:H)

Angle of the embankment with the horizontal ( $\beta$ )= $\tan^{-1}\left(\frac{1}{2}\right) = 26.6^\circ$

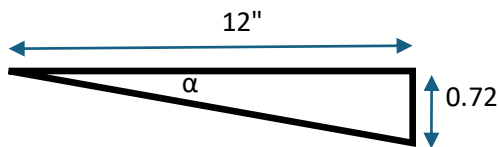


#### Angle of the Cross-fall

cross-slope=6%

0.72" of fall per horizontal 12" of road width

Angle of the Cross-fall with the horizontal ( $\alpha$ )=  $\tan^{-1}\left(\frac{0.72}{12}\right) = 3.43^\circ$



#### Concrete Calculations

For a stretch of 100 m of the road;

Volume of pedestrian walkaway ( $V_1$ ) = 7 m<sup>3</sup>

Volume of base of the green wall ( $V_2$ ) = 314.64 m<sup>3</sup>

Volume of stairways ( $V_3$ ) = 0.1632 m<sup>3</sup>

Volume of culvert drain ( $V_4$ ) = 0.048+14.86 = 14.91 m<sup>3</sup>

Volume of the bridge (V5) = 0.058 m<sup>3</sup>

Volume of the side drains (V6) = 0.024 m<sup>3</sup>

Total concrete volume = V1+V2+V3+V4+V5+V6 = 336.8 m<sup>3</sup>

- To make 1 cubic meter of concrete, you will need: 7 bags cement + 0.70 cubic meters sand + 0.70 cubic meters stone (*Build It - What Is the Correct Batch Ratio for Mixing Concrete?*, n.d.).

Hence, for the total concrete volume;

Number of concrete bags = 7 x 336.8 = **2358 bags**

Volume of sand = 0.70 x 336.8 = **235.76 m<sup>3</sup>**

Volume of stones = 0.70 x 336.8 = **235.76 m<sup>3</sup>**

Except for V1 and V6, the rest of the concrete volumes need to be reinforced with steel.

- Following the rule of thumb steel weight in footing for one cubic meter of concrete is 1% of the wet concrete weight (*Studocu, n.d.*).

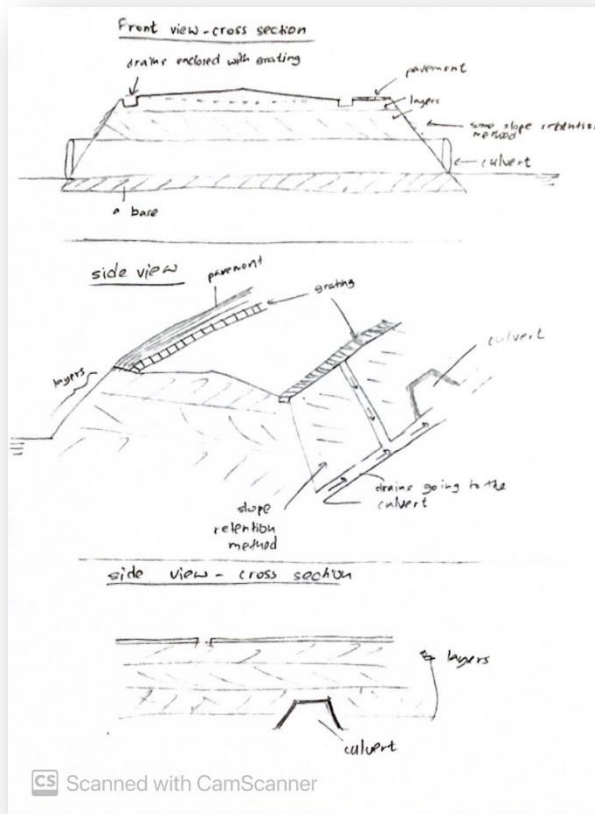
Hence; steel weight per one cubic meter of concrete = 24 kg

Weight of steel required = (336.8 - (7+0.024)) x 24 = **7914.6 kg**

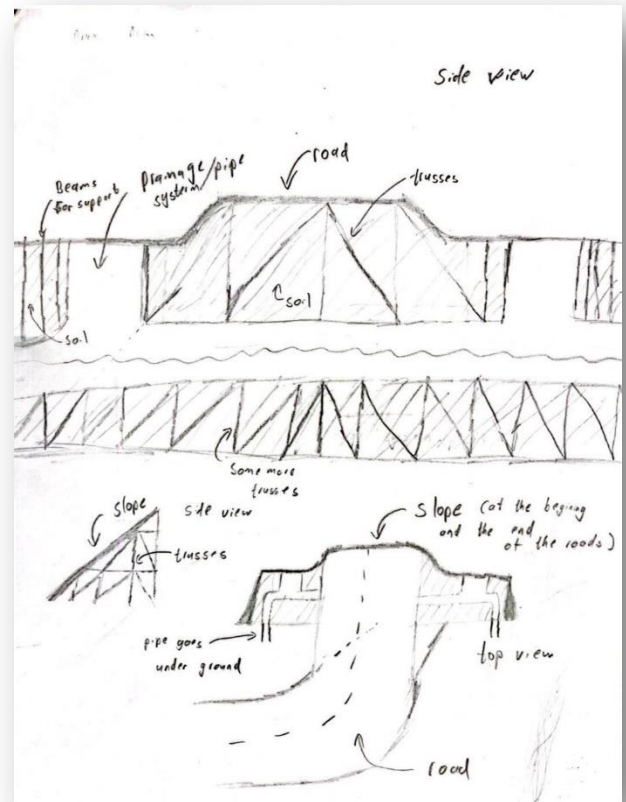


## Appendix B - Alternative Conceptual Designs

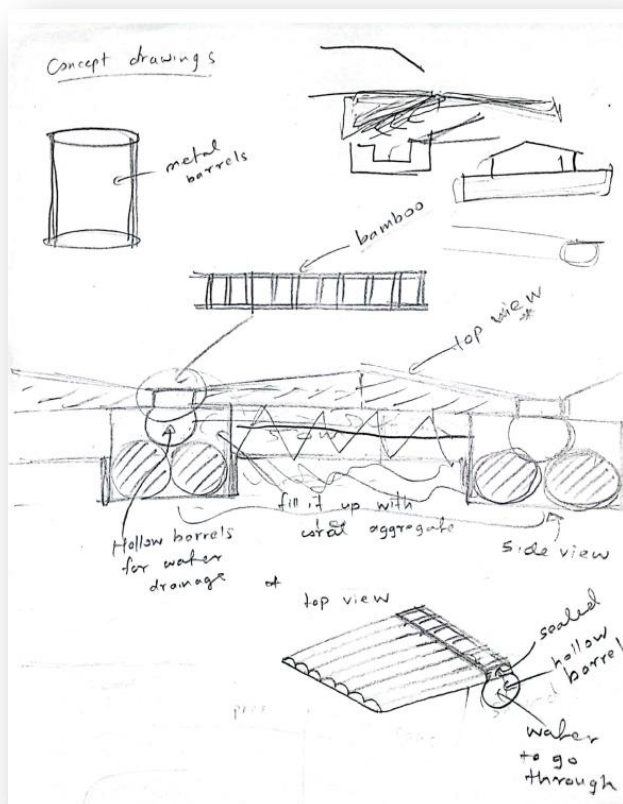
Chiranga Karunaratne's conceptual design



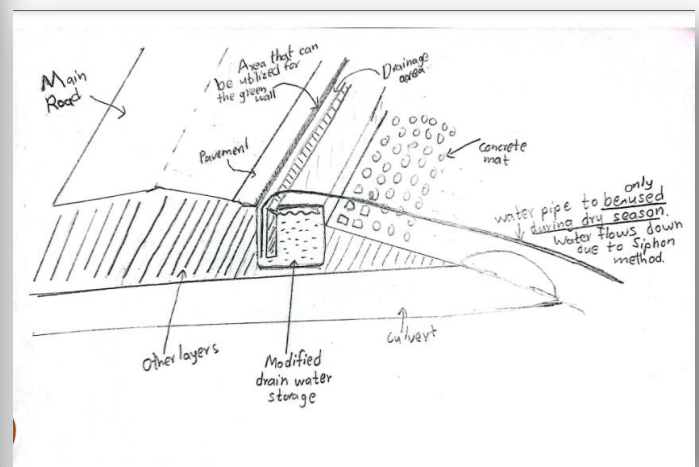
Yameen Shariz's conceptual design



Hashin Maharambe's conceptual design



Rovindu Tharin's conceptual design



## Appendix C – Table of Contributions from each member

| Name                 | Sections contributed to the report                                                                                                                                                                                                                                                                                                          |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rovindu Tharin       | <ul style="list-style-type: none"> <li>• Project Background – Climate</li> <li>• Stakeholders – Investors/Partner organization</li> <li>• Specifications</li> <li>• Design Criteria and Constraints - Sustainability</li> <li>• Tender Evaluation Criteria</li> <li>• Executive Summary</li> <li>• Appendix C</li> </ul>                    |
| Hashin Maharambage   | <ul style="list-style-type: none"> <li>• Project Background – Political Background &amp; Geography and Environment</li> <li>• Stakeholders – Constructors and Suppliers</li> <li>• Cover Page</li> <li>• Specifications</li> <li>• Design Criteria and Constraints- Durability</li> <li>• Introduction</li> <li>• Model Drawings</li> </ul> |
| Chiranga Karunaratne | <ul style="list-style-type: none"> <li>• Project Background – Economic want to the project</li> <li>• Stakeholders - Local Government</li> <li>• Design Criteria and Constraints - Safety</li> <li>• Conclusion</li> <li>• Concept Solution</li> <li>• Concept Drawings</li> <li>• Appendix A</li> <li>• Appendix D</li> </ul>              |
| Yameen Shariz        | <ul style="list-style-type: none"> <li>• Project Background – Culture, Demographics &amp; Education</li> <li>• Stakeholders – Customer and the community</li> <li>• Specifications</li> <li>• Design Criteria and Constraints - Materials</li> <li>• Tender Evaluation Criteria</li> </ul>                                                  |

|  |                                                                                          |
|--|------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"><li>• Executive Summary</li><li>• Appendix B</li></ul> |
|--|------------------------------------------------------------------------------------------|

## Appendix D - Approval for Dimension Constraint Relaxation

Re: INDE1001 - Urgent - Request for Approval to Relax Dimension Constraint

Omila Fernando <omila.f@slit.lk>

Sat 4/13/2024 5:46 PM

To: Mahadura Rovindu Tharin De Silva (Student) <m.desilva12@student.curtin.edu.au>

Cc: Nimal Visahan <nimal.v@slit.lk>

Yes, you may proceed to relax the height to not less than 150 mm

Best regards,  
Omila Fernando

Sent from [Outlook for Android](#)

---

**From:** Mahadura Rovindu Tharin De Silva (Student) <m.desilva12@student.curtin.edu.au>

**Sent:** Saturday, April 13, 2024 2:42:03 PM

**To:** Omila Fernando <omila.f@slit.lk>

**Cc:** Nimal Visahan <nimal.v@slit.lk>

**Subject:** INDE1001 - Urgent - Request for Approval to Relax Dimension Constraint

Some people who received this message don't often get email from m.desilva12@student.curtin.edu.au. [Learn why this is important](#)

**[EXTERNAL EMAIL] This email has been received from an external source – please review before actioning, clicking on links, or opening attachments.**

Greetings Sir,

I am Rovindu Tharin, the INDE1001 group "Yellow Squad" leader. Our group is doing the project on the local road access improvement in Saibai Island.

In our representative model, the total length in x dimension is 453.8 mm, and in y, it is 200 mm. However, building our representative model length in z dimension above 200 mm is impractical with regard to the criteria mentioned in section 2.9 of the design package. The total length in the z-dimension of our representative model is 155 mm (which is still above the limit of the possible relaxed constraint).

Below I have mentioned the reasons that we find it impractical for our specific model. Please note that the two related drawings are attached to this email as PDFs.  
(The total lengths in all x, y and z dimensions are included in "Overall Lengths.pdf" and a drawing with 2 views which shows most of the main parts are included in "Concept Drawing 2.pdf")

Reasons:

In our model, the total length in x dimension is 453.8 mm (as shown in "Overall Lengths.pdf") including the Culvert Drainage system (as shown in view 4 of "Concept Drawings 2.pdf") which is a vital component of our design solution to direct flood water to sewers or the lagoon. It maintains safety and ensures the proper functioning of our design in order to meet our design criteria. To achieve a 200 mm length in z-dimension, either we will have to increase the length of the whole model (then the x-dimension length will go beyond 500 mm which will conflict with the criteria) or we will have to increase the height of the street lamp (which is impractical since according to our engineering scale, such street lamps of big heights cannot be practically implemented.) or will have to increase the height of the

structure(which will then again conflict with our low-cost criteria) . Thus, in all of our options, it is impractical to have a length of 200mm in z dimension while protecting the maximum length below 500mm.

Therefore, considering the above-mentioned details and our limited time constraint to make changes , in accordance with the second point mentioned under section "2.9 Representative Model" of the design package, I kindly request to grant approval in order to relax this minimum dimension constraint of 200mm to 150 mm.

The earliest response is highly appreciated.  
Thank you for your time and consideration.

Kind Regards.  
Rovindu Tharin.  
Curtin ID : 22188221  
Leader, INDE1001 Group : Yellow Squad

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